Taming Iterative Grinding Attacks on Blockchain Beacons

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Abstract. Random beacons play a critical role in blockchain protocols by providing publicly verifiable, unpredictable randomness essential for secure assignment of protocol roles such as block producers and committee membership. In the interest of efficiency, many deployed blockchains adopt beacon algorithms that suffer from *grinding*: an adversarial attack in which a party exploits freedom given by the protocol to bias the outcome of the random beacon by resampling it several times and picking the most desirable outcome. To compound the problem, beacons often operate in an iterative manner, where the beacon output produced during one protocol epoch serves as the random seed for the beacon's invocation in the next epoch. This amplifies the security threat, as such attacks may then aggregate their power over many epochs.

In this article, we formulate a generic framework for information-theoretic analysis of grinding in iterated randomness beacons. We define the *natural grinding capacity* of a beacon, intuitively corresponding to the amount of grinding it allows with a uniformly random seed. We then prove that sufficiently strong tail bounds on this quantity can be transformed into a guarantee on smooth min-entropy of the iterated beacon's output, even conditioned on all past outputs and irrespective of the inner workings of the beacon. Such min-entropy guarantees can immediately be translated into corresponding statements about various applications of the beacon to committee selection, incentives, or underlying protocol security.

Our main technical result concerns conventional longest-chain protocols, where we establish that the combinatorial structure of the forest of longest chains can be leveraged to control grinding. Instantiating the generic framework with these grinding upper bounds, we establish that the randomness beacon of the Ouroboros Praos protocol is secure against adversaries controlling up to about 12% of stake—even without any assumptions bounding the adversarial computational power invested into grinding. This is a qualitatively new guarantee for the protocol.

1 Introduction

Distributed randomness generation has long played a central role in distributed computing, owing both to the utility of shared randomness in algorithm design and to the rich mathematical structure of the area. In recent years, the deployment of large-scale ledger consensus protocols—especially those adopting the proof-of-stake mechanism—has brought renewed urgency to this research program, driving demand for randomness generation protocols that are not only cryptographically sound but also scalable, resilient to adaptive adversaries, and secure under composition.

A common architecture arising in these consensus algorithms—such as those underlying the blockchains Ethereum, Cardano, Polkadot, and others—organizes the protocol into a sequence of "epochs." Each epoch is governed by a randomness nonce that seeds core consensus operations—such as leader election and committee sampling—while also supporting higher-level functionalities, such as smart contract execution, rewards distribution, layer-2 protocols, and zero-knowledge proofs. Implicit in this design is the demand that each epoch also determines the nonce for the following epoch. This poses a unique security challenge: the nonce generation procedure must resist adversarial influence under prolonged attacks that accumulate bias over multiple epochs.

The protocols adopted by these blockchains to solve this task differ, but similar efficiency and security considerations have led most of them to adopt solutions following a single paradigm. This calls for each eligible participant to broadcast an (ostensibly) random value, after which the blockchain's consensus protocol is leveraged to agree on a subset $\mathcal{S}$ of these values. The final beacon value is then determined by hashing the selected values together (that is, $\mathsf{H}(\mathcal{S})$ for a canonical representation of $\mathcal{S}$). An early innovation in this space calls for each participant to produce its contribution via a verifiable random function (VRF [5]), preventing direct influence of parties over their contributions.

However, even if the hash function $\mathsf{H}(\cdot)$ used to compute the resulting beacon output itself is modeled as a random oracle, this approach does not necessarily produce perfectly uniform randomness. The reason is that an adversary participating in the consensus protocol might attempt to steer its outcome to affect the resulting set $\mathcal{S}$ toward a favorable value. Notice that since the selected values $\mathcal{S}$ are combined into the beacon's output by a public deterministic function, the adversary can locally evaluate the beacon outcome for any candidate set $\mathcal{S}$ and choose among them—an attack commonly referred to as *grinding*. Note that many consensus protocols allow for such limited adversarial influence over the choice of $\mathcal{S}$, as it does not necessarily conflict with their main design goals of achieving safety and liveness.

One method to address this concern in formal security analyses, taken for instance in [4], is to rely on an upper bound on the computational power of the adversary, effectively limiting the number of evaluations of the function $\mathsf{H}(\cdot)$ that the adversary is capable of carrying out during the "opportunity window" before the set $\mathcal{S}$ is fixed by the consensus protocol. This approach—while sufficient to yield protocol security and additionally consistent with physical reality where computational power is limited—partially defeats the purpose of designing a proof-of-stake protocol, as security of the resulting protocol must then rely on additional computational assumptions (beyond the stake-based assumptions required for the consensus lottery and, of course, the security of the underlying cryptographic primitives).

To appreciate the threat of grinding on these protocols when it is simply circumscribed by $\mathsf{H}()$ query complexity as discussed above, it is instructive to express their resulting insecurity as a sum of two terms, ϵ_C and ϵ_Π: (i.) ϵ_C reflects the insecurity of the underlying cryptographic primitives, and decays as a function of a cryptographic security parameter λ_C; (ii.) ϵ_Π reflects the structural insecurity of the protocol itself, and decays as a function of an independent structural security parameter λ_Π. While λ_C can be adjusted at the expense of a modest increase in computational complexity in order to achieve desirable ϵ_C, the security parameter λ_Π is tethered to important protocol features—such as settlement times—and cannot be adjusted freely. Grinding introduces an additional multiplicative factor affecting ϵ_Π, scaling with the assumed bound on total query complexity. This provides strong motivation for a mechanism to control grinding that does not require compensation via adjusting λ_Π.

A general challenge facing approaches to grinding mitigation is the concern that the effects of grinding may be compounded over several epochs. The main idea of such an attack is to leverage grinding in some single epoch e to arrive at a beacon output that, via its effect on the consensus protocol in the subsequent epoch $e+1$, puts the adversary in an even more favorable position to perform grinding in epoch $e+1$, further improving their grinding capabilities in epoch $e+2$, and so on. Without taming the grinding capability globally (such as by the computational assumption above), it is a priori unclear whether this iterative process could even lead to an eventual full compromise of the beacon and, by extension, the consensus protocol—a disastrous outcome. We refer to this attack as *iterated grinding*, which is the primary object of study in this work.

The above discussion motivates the following general questions about grinding attacks in the context of consensus protocols with iterated epoch structure:

How does adversarial leverage grow during iterated grinding attacks?
Is it globally bounded or could it eventually break the protocol's security?
What entropy guarantees does the beacon provide after several iterations?

1.1 Our Contributions

We make the following contributions in this article:

A general framework for iterative grinding analysis. We pick up a thread of research begun in [12] and introduce a general information-theoretic framework for studying the effects of iterative grinding on blockchain beacons. In our model, any beacon that allows the adversary to choose from a set of independent uniform output candidates is abstracted as a formal object we call a *grindable beacon*, parameterized by its *natural grinding capacity* and *failure rate*; its output is then described using the concept of *optative random variables*. We then establish a formal mechanism for composing such beacons that models practice.

The principal technical fulcrum of this approach is a composition theorem (Theorem 1) that converts tail bounds on a beacon's grinding capacity into high smooth min-entropy guarantees for its iterated outputs—even conditioned on all prior epochs. This abstraction is agnostic to the underlying consensus mechanism and could be applied equally to Ethereum's RANDAO, Cardano's Praos beacon, Polkadot, Avalanche, Snow White [3] or any other protocol using such an iterated randomness beacon. The theorem enables rigorous reasoning about both protocol security and beacon entropy loss under sustained grinding. Backed by our composition theorem, we introduce the notion of a beacon being *secure under composition*—a property that exactly allows us to invoke this theorem to conclude that the beacon maintains negligible failure probability and a linear fraction of min-entropy also under polynomial sequential iteration.

Selfish-mixing attacks. As a preliminary application of our framework, we revisit the well-studied selfish mixing attack [16,1], where an attacker strategically misses block-creation opportunities toward the end of an epoch in order to influence the lottery outcome of the subsequent epoch. We formalize this strategy within our abstraction by defining a grindable beacon that captures this adversarial behavior, and show that the beacon remains secure under composition provided the adversary controls a minority of stake.

Controlling grinding in longest-chain protocols: the Cardano Praos beacon. We establish tail bounds on the grinding power of conventional longest-chain rule protocols that (repeatedly) determine beacon values as a function of the chain settled at particular "strike times." This permits us to apply the general framework above to achieve control of the min-entropy of resulting beacon values. In particular, we study the Ouroboros Praos beacon deployed in Cardano and obtain the first proof that Praos remains secure against adversaries holding up to $\approx 12\%$ of the stake *without any computational assumption limiting their grinding power*—thereby removing a key assumption from the original Praos analysis [4]. Moreover, under a natural *density assumption* empirically satisfied by Cardano—namely, that the finalized chain contains at least its expected number of honest-party blocks—we strengthen this bound to adversaries controlling up to $\approx 15\%$ of the stake. These results deliver a qualitatively novel security guarantee for Cardano's deployed beacon and illustrate the practical impact of the framework.

1.2 Technical Overview

To capture grindable beacons, we leverage the notion of a *g-optative random variable* [12]: intuitively, this is a random variable describing a value chosen from a set of g uniformly random, freshly sampled random variables, the choice being carried out based on full knowledge of the sampling outcomes. Notice that this is exactly the situation of a grinding adversary, motivating the technical definition.

We then define a *grindable beacon* $\mathbf{B}$ as an abstract and relatively simple mathematical object: a family of pairs $(\mathcal{G}_x, \epsilon_x)_{x \in \{0,1\}^\lambda}$ indexed by a seed $x \in \{0,1\}^\lambda$ where, for each x,

– $\mathcal{G}_x$ is the distribution of the number of "grinding options" an adversary will be able to choose from in an execution seeded by x, and
– ϵ_x is the probability that the protocol's execution seeded by x (and hence also the respective beacon execution) fails.

It turns out that the important characteristics of a grindable beacon are its *natural grinding capacity* $\mathcal{G}(\mathbf{B})$ and *natural failure rate* $\epsilon(\mathbf{B})$, which, roughly speaking, correspond to the above-described two quantities under a uniformly random input seed $X \xleftarrow{\$} \{0,1\}^\lambda$.

Formalizing an approach implicit in [12], we introduce a *grinding game* which describes how the minimalistic description of a grindable beacon can be used to derive the beacon's output: unless the input symbol x indicates a failure of a previous beacon (having sequential composition in mind) and the current invocation does not fail either (with probability $1-\epsilon_x$), the game samples the amount of potential grinding $G \leftarrow \mathcal{G}_x$ and samples a G-optative random variable, in which the choice among the underlying uniform values is left to the adversary; this represents the beacon's output. It is then easy to formalize how such beacon invocations are sequentially composed.

We then prove the main composition theorem (Theorem 1), which shows that $\epsilon(\mathbf{B})$ and a tail bound on $\mathcal{G}(\mathbf{B})$ can together be translated into an upper bound on the failure of the iterated execution of the beacon, as well as a lower bound on the min-entropy of the beacon's output after the iterated execution, assuming the failure has not occurred. More concretely, any tail bound on the natural grinding capacity of the form $\Pr[\mathcal{G}(\mathbf{B}) > \mathit{\Gamma}] \leq \gamma$ implies that after k epochs the overall failure probability is bounded by $k\mathit{\Gamma}(\gamma + \epsilon(\mathbf{B}))$, while the final output, conditioned on survival, retains roughly $\lambda - \log_2 \mathit{\Gamma}$ bits of min-entropy (here λ denotes the output length).

Equipped with this general framework, we proceed to our main technical effort: controlling the grinding capacity of longest-chain rule protocols under the natural assumption that the beacon value is determined by the longest chain at a particular "strike point" in the protocol. This is exactly the setting of interest for the Ouroboros Praos beacon deployed in the Cardano blockchain. After setting the stage by casting this beacon as an abstract grindable beacon, the challenge is to establish strong tail bounds on its natural grinding capacity.

While abstract longest-chain-rule protocols have received significant analytic attention in the literature, existing techniques do not provide adequate control of the "breadth" of the blocktree of longest chains, and obtaining a tail bound on the natural grinding capacity hence requires a new technical approach. A delicate issue in the analysis concerns the trade-off between the *probability* of "long forks" in the longest chain and the *size* of the associated family of longest chains (which controls the number of grinding attempts): in particular, while the probability of long forks is known to decay exponentially in the length of the fork, the total number of such alternate chains can grow exponentially in the length. We approach this by controlling the $(1+\delta)$-moment, for appropriately chosen δ, which demands fine control of these blockchain breadth statistics. This leads to our main result, Theorem 3, showing that for sufficiently weak adversaries, single-epoch security of Ouroboros Praos with perfect randomness implies multi-epoch security of that protocol with its randomness produced by the iterated beacon, as well as quantifying the min-entropy of the final output of the beacon.

Interestingly, our analysis highlights that a more favorable bound on the relevant moment can be obtained under the additional (observable) fact that the chain produced by the protocol is sufficiently dense in the sense that its block density corresponds at least to what is *expected* based on honest participation. We empirically verify that this assumption is satisfied in the case of Cardano, and show that under this assumption, the previously obtained security guarantees can be established also for stronger adversaries (Theorem 4).

Finally, note that for simplicity, our model of Praos execution adopts a lockstep synchronous networking assumption, although the protocol has been fully analyzed in a more realistic $\mathit{\Delta}$-delay model. We explain in Appendix 6 that our analysis could be reproduced in the $\mathit{\Delta}$-delay model and would lead to qualitatively similar results.

1.3 Related Work

In the context of grinding attacks against iterated blockchain beacons, existing literature has focused primarily on the Algorand and Ethereum blockchains.

In the case of Algorand, [7,6] study the attack where an adversary controlling several parties—all among the top candidates for a block proposer in a given round—can decide which of the controlled parties actually produces a block depending on the favorability of the resulting beacon outcome. The paper [7] fully analyzes a simple 1-lookahead strategy that always outperforms honest execution, and describes a Markov Decision Process (MDP) that characterizes the optimal strategy. Subsequent work [6] then provides efficient computational methods to derive tighter estimates of the optimal attack's advantage.

For Ethereum, the initial focus was on so-called *selfish-mixing* attacks [16] mentioned above. Alpturer and Weinberg [1] determine optimal selfish mixing strategies using MDPs. A broader class of grinding attacks, involving also forking the chain, was analyzed in [14]. Finally, [15] further broadened the set of considered attacks, while at the same time bringing attention to another potential adversarial goal: becoming a leader for a particular future slot.

From an analytic perspective, the work closest to ours is [12], which established the notion of g-optative random variables in order to provide an iterative analysis of a bespoke randomness beacon algorithm (similar in spirit to the Algorand beacon). Indeed, the generic grindable beacon framework we articulate in Section 2.2 of this paper formalizes a concrete approach implicit in the proof of that paper.

2 Modeling Iterative Blockchain Beacons

Basic Notation. Let $\mathbb{N} = \{1, 2, \ldots\}$ denote the set of natural numbers and for each $n \in \mathbb{N}$, let $[n] = \{1, 2, \ldots, n\}$. As is standard, we use $\ln(\cdot)$ and $\log(\cdot)$ for natural and base-2 logarithms, respectively. We denote sampling X from a set $\mathcal{X}$ uniformly at random by $X \xleftarrow{\$} \mathcal{X}$.

For a random variable X with support $\mathcal{X}$, we denote by $H_\infty(X)$ and $H_\infty^{(\epsilon)}(X)$ its *min-entropy* and *ϵ-smooth min-entropy*, respectively:

$$H_\infty(X) = -\log\left(\max_{x\in\mathcal{X}} \Pr[X = x]\right) \qquad \text{and} \qquad H_\infty^{(\epsilon)}(X) = \max_{Y\sim_\epsilon X} H_\infty(Y)\,,$$

where this maximum is taken over all random variables Y for which the statistical distance between X and Y is no more than ϵ.

We denote by λ the security parameter, and throughout the paper we assume that all beacons produce λ-bit outputs.

Intuition. The next sequence of definitions capture the notion of a randomness beacon. Informally, a λ-beacon $\mathbf{B}$ is a process defined in terms of a random oracle $f : \{0,1\}^* \to \{0,1\}^\lambda$ with the following properties:

1. The beacon is seeded with a random value $x \in \{0,1\}^\lambda$.
2. A distributed computation takes place—which may involve adversarial corruption of participating parties—in which a collection of parties, each provided with x and (oracle access to) f, determine a value v.
3. The output of the beacon is $f(v) \in \{0,1\}^\lambda$.

In light of the fact that a random function is applied to v in step (3), it is natural to expect that the output of the beacon provides some guarantee of randomness. In practice, we typically find that the value v is under partial adversarial control in the sense that the protocol effectively identifies a collection of random values $\{v_1, \ldots, v_g\}$ from which the adversary may select the final v; as the adversary may have queried $f(v_i)$ for each element of this "candidate set," the adversary can certainly exert some control over the final result.

Ultimately, we will be interested in the behavior of a beacon under iteration $x_0 \to x_1 \to x_2 \to \cdots$, where x_0 is drawn uniformly in $\{0,1\}^\lambda$, and each instance of the beacon is seeded with the result of the previous instance. The goal, intuitively, is to guarantee that with high probability each x_i has high min-entropy, even when conditioned on prior x_j.

Motivating smooth min-entropy as the figure of merit. To explain why min-entropy is a satisfactory quantity to control, we remark that if X is a random variable taking values in $\{0,1\}^\lambda$ with min-entropy $\lambda - \nu$ then for any event $B \subset \{0,1\}^\lambda$

$$\Pr[X \in B] \leq 2^\nu \cdot \Pr[U \in B] \tag{1}$$

where U is uniform on $\{0,1\}^\lambda$. In particular, imagining B as a "bad" event in the context of a randomized distributed algorithm, the probability that B occurs under the (imperfect) beacon value X is no more than 2^ν times the probability that the event would arise under a perfectly uniform beacon value. Of course, in our setting, where the beacon is responsible for generating a sequence of potentially correlated values $X_1, X_2, \ldots$ we must insist on control of min-entropy even when *conditioned on all previous beacon values.* In this case, for any sequence of bad events $B_1, \ldots, B_n$, where B_i is an event determined by the ith beacon value, $\Pr[X_i \in B_i] \leq 2^\nu \Pr[U \in B_i]$ and it follows that the probability that any bad event occurs is no more than

$$2^\nu \sum_i \Pr[U \in B_i] \tag{2}$$

(where, as above, $\Pr[U \in B_i]$ is the probability that the event occurs with an ideal, uniform beacon value). This is exactly the framework in which randomness nonces arise in proof-of-stake ledger protocols such as Ouroboros Praos. In particular, defining B_i to be the event that a consensus consistency or liveness failure occurred in the ith epoch, this controls the probability of such a failure over the lifetime of a protocol driven by protocol nonces with bounded (conditional) min-entropy.

Looking ahead, for technical reasons we work with a slightly more flexible notion than min-entropy: *smooth min-entropy*. The reason for this is that the randomness generation algorithms in typical ledger protocols build on the underlying consensus layer and so may have a small probability of a "complete failure" corresponding to, e.g., circumstances where the underlying ledger fails to achieve safety or liveness. Of course, this could be directly reflected using min-entropy, but it is useful to separate this source of infrastructural error from the imperfection of the beacon (measured by min-entropy) because in many cases we can prove much stronger bounds on the distortion this way. Specifically, we will prove that—except for a failure event E_t corresponding to infrastructural error—the conditional min-entropy of each X_t is $\lambda - \nu$ for each t. There is a corresponding "generic" probability upper bound, analogous to Equation (1), that applies in this case: if $\Pr[E_t] \leq \delta$, then for any bad event B, $\Pr[X_i \in B_i] \leq \delta + 2^\nu \Pr[U \in B_i]$. This leads to an additional additive term of δn in the probability corresponding to (2) that a bad event occurs "in any round" of a n-round protocol. The reader can observe the appearance of these two error terms in the main beacon composition statement (Theorem 1).

Consequences for committee sampling and rewards. Aside from the fundamental effect of grinding on consensus failures, nonces may also determine reward assignment or membership on committees relevant for the protocol or layer-2 functionalities. In these settings, where one wishes to estimate the fraction of a committee or the fraction of rewards assigned to a particular (say, adversarial) collection of parties, a conventional Chernoff bound is typically used to control the unlikely tails of the distribution. Roughly, in both cases one studies a random variable F—representing the fraction of incentives or committee seats assigned to the collection of parties—for which $\Pr[|F - \mathbb{E}[F]| \geq \lambda] \leq \exp(-\theta(\lambda^2) \cdot n)$, where n is the total number of committee seats or parties to which incentives are assigned, and the constant hidden in the θ notation may depend on the expectation $\mathbb{E}[F]$. Then, in case such selection is carried out using a nonce with entropy deficit ν (that is, the λ-bit nonce has min-entropy $\lambda - \nu$), this probability is simply amended to $\exp(-\theta(\lambda^2)n + \nu \ln(2))$. Thus the very same apparatus can be used to control the behavior of such group assignment mechanisms under nonces with controlled min-entropy.

2.1 Optative Random Variables

The above picture motivates identification of the following class of random variables. Following [12] we call a random variable *g-optative* if it can be expressed as the result of an "adversary" selecting from a population of g independent and uniformly random values.

Definition 1 (g-optativity [12]). *We say that $C\colon (\{0,1\}^\lambda)^g \to \{0,1\}^\lambda$ is a g-*choice function *if it satisfies $C(u_1, \ldots, u_g) \in \{u_1, \ldots, u_g\}$ for any tuple of inputs $(u_1, \ldots, u_g) \in (\{0,1\}^\lambda)^g$. A random variable X taking values in $\{0,1\}^\lambda$ is called g-*optative *if there is a g-choice function $C\colon (\{0,1\}^\lambda)^g \to \{0,1\}^\lambda$ such that $X = C(U_1, \ldots, U_g)$, where $U_1, \ldots, U_g$ are uniform, independent random variables in $\{0,1\}^\lambda$.*

The following observations follow easily from the definition of g-optativity.

Lemma 1. *Let $X = C(U_1, \ldots, U_g)$ be a g-optative random variable over $\{0,1\}^\lambda$.*

(i) X is g'-optative for any $g' \geq g$.

The Grinding Game $\mathsf{GG}^{\mathbf{B},\mathcal{A}}(x)$

(G1) **sample** $G \leftarrow \mathcal{G}_x$
(G2) Sample a G-optative random variable Y:
  (a) $\mathcal{A}(x)$ outputs a G-choice function C
  (b) $Y = C(U_1, \ldots, U_G)$ for fresh independent uniform U_i
(G3) **with probability** ϵ_x **return** $\perp$
(G4) **if** $x = \perp$ **return** $\perp$
(G5) **return** Y

Fig. 1. Grinding game: Evaluating a beacon $\mathbf{B} = (\mathcal{G}_x, \epsilon_x)_{x\in\{0,1\}^\lambda}$ on input $x \in \{0,1\}^\lambda \cup \{\perp\}$ in presence of an adversary $\mathcal{A}$.

(ii) For any $x \in \{0,1\}^\lambda$, $\Pr[X = x] \le \Pr[x \in \{U_1, \ldots, U_g\}] \le g/2^\lambda$.
(iii) The min-entropy of X is upper-bounded by $\lambda - \log(g)$, and this remains true even when conditioning on any event E independent of $U_1, \ldots, U_g$:

$$H_\infty(X \mid E) \ge \lambda - \log(g)\,.$$

Proof. Claims (i) and (ii) follow directly from the definition, claim (iii) is a consequence of (ii). □

The notion of g-optativity can be naturally formulated in the general setting where g is a random variable: this calls for an associated random variable C so that, conditioned on any particular value g_0 for g, X is determined by the g_0-choice function C (applied to independent u_i).

Given that the min-entropy of optative random variables follows Lemma 1(iii), the following lemma will be applicable to such variables; its proof is immediate from the definition of min-entropy.

Lemma 2. *Let $f : \{0,1\}^\lambda \to \mathbb{R}$ be a non-negative function. Let E_f denote the quantity $2^{-\lambda}\sum_{x\in\{0,1\}^\lambda} f(x)$ equal to the expected value of f taken over the uniform distribution on $\{0,1\}^\lambda$. Let X be a random variable taking values in $\{0,1\}^\lambda$ with $H_\infty(X) \ge \lambda - \kappa$; then $\mathbb{E}[f(X)] \le 2^\kappa \cdot E_f$. If $f(X)$ is the indicator random variable for an event $A \subset \{0,1\}^\lambda$, it follows that $\Pr[X \in A] \le 2^\kappa \cdot P_A$, where $P_A = |A|/2^\lambda$ is the probability that a uniformly selected element of $\{0,1\}^\lambda$ lies in A.*

2.2 Grindable Randomness Beacons

We here set down a formal, generic approach to iterative grinding analysis inspired by the analysis in [12] that considered an iterated sequence of grinding tail bounds to analyze a particular beacon. The approach has the advantage that it applies to an arbitrary beacon, independent of internal structure, so long as the ultimate beacon value is g-optative (that is, guaranteed to be an element of a set of independent, uniform random values determined—perhaps implicitly—during the procedure). The analysis then depends on the random variable given by the size of this set of potential values; in order to best model practice, the model is slightly amplified to carry an infrastructural failure term, as discussed above.

Definition 2 (Grindable Beacon). *For a fixed security parameter $\lambda \in \mathbb{N}$, a* grindable beacon *(or* beacon *for short) is a family of pairs $\mathbf{B} = (\mathcal{G}_x, \epsilon_x)_{x\in\{0,1\}^\lambda}$ where, for each $x \in \{0,1\}^\lambda$, $\mathcal{G}_x$ is a distribution over $\mathbb{N}$ and $\epsilon_x \in [0,1]$.*

The semantics of the above minimalistic description is given by the *grinding game* $\mathsf{GG}^{\mathbf{B},\mathcal{A}}(x)$ of Fig. 1, describing the evaluation of a beacon $\mathbf{B}$ on input x in the presence of an adversary $\mathcal{A}$. In a nutshell, given input x, the beacon produces a fresh G-optative beacon output $Y \in \{0,1\}^\lambda$ for G distributed according to $\mathcal{G}_x$, unless it fails (e.g., due to an underlying protocol error), which happens with probability ϵ_x and results in a special output symbol $\perp$. Since we will be interested in sequential composition of beacons, we also treat $\perp$ as a legal input to the evaluation game with the understanding that in this case the game simply returns $\perp$.

We will use the notational shorthand $x \triangleright^{\mathcal{A}} \mathbf{B}$ to refer to the output $Y \in \{0,1\}^\lambda \cup \{\perp\}$ of the experiment $\mathsf{GG}^{\mathbf{B},\mathcal{A}}(x)$. We occasionally overload this notation and refer by $x \triangleright^{\mathcal{A}} \mathbf{B}$ to the respective random experiment defining the random variable $Y \leftarrow \mathsf{GG}^{\mathbf{B},\mathcal{A}}(x)$, it should always be clear from the context whether we refer to the experiment or its output. Whenever the adversary $\mathcal{A}$ is clear from the context we simply write $x \triangleright \mathbf{B}$. We will presently extend this notation to reflect the result of composing a sequence of beacons, for which we will adopt the notation $x \triangleright^{\mathcal{A}} \mathbf{B}_1 \triangleright^{\mathcal{A}} \mathbf{B}_2 \triangleright^{\mathcal{A}} \cdots$; see Def. 4 below.

Definition 3 (Natural grinding capacity and failure rate). *Let* $\mathbf{B} = (\mathcal{G}_x, \epsilon_x)_{x \in \{0,1\}^\lambda}$ *be a beacon. The* natural grinding capacity *of* $\mathbf{B}$*, denoted* $\mathcal{G}(\mathbf{B})$*, is the distribution*

$$\mathcal{G}(\mathbf{B}) \triangleq \mathcal{G}_X \quad \textit{for a uniformly random} \quad X \stackrel{\$}{\leftarrow} \{0,1\}^\lambda .$$

The natural failure rate *of* $\mathbf{B}$*, denoted* $\epsilon(\mathbf{B})$*, is the quantity*

$$\epsilon(\mathbf{B}) \triangleq \sum_{x \in \{0,1\}^\lambda} \epsilon_x / 2^\lambda .$$

Intuitively, $\mathcal{G}(\mathbf{B})$ is the distribution of the "grinding capacity" (i.e., of the random variable describing the beacon output's optativity) under a uniformly random input $X \stackrel{\$}{\leftarrow} \{0,1\}^\lambda$, while $\epsilon(\mathbf{B})$ is the probability of the beacon $\mathbf{B}$ failing when invoked on a uniformly random input.

2.3 Sequential Beacon Composition

We now describe the natural way to sequentially compose beacon evaluation. We describe the general case for completeness; looking ahead, our subsequent results will consider the most common case of k-wise sequential composition of *the same* beacon.

Definition 4 (Evaluation of composed beacons). *Fix a security parameter* $\lambda \in \mathbb{N}$ *and an integer* $k \in \mathbb{N}$*. Given* k *beacons* $\mathbf{B}_1, \mathbf{B}_2, \ldots, \mathbf{B}_k$ *and an adversary* $\mathcal{A}$*, the* (sequential) evaluation of $\mathbf{B}_1, \mathbf{B}_2, \ldots, \mathbf{B}_k$ at $x \in \{0,1\}^\lambda \cup \{\perp\}$*, denoted* $x \triangleright^{\mathcal{A}} \mathbf{B}_1 \triangleright^{\mathcal{A}} \cdots \triangleright^{\mathcal{A}} \mathbf{B}_k$*, refers to the output of the random experiment*

$$Y \leftarrow \mathsf{GG}^{\mathbf{B}_k,\mathcal{A}}\left(\cdots \mathsf{GG}^{\mathbf{B}_2,\mathcal{A}}\left(\mathsf{GG}^{\mathbf{B}_1,\mathcal{A}}(x)\right)\right) , \tag{3}$$

where the adversary $\mathcal{A}$ *is allowed to maintain state across the* k *sequentially executed instances of the game* GG *(we keep this implicit in the notation). We extend this notation to the intermediate results of the evaluation experiment, letting* $x \triangleright^{\mathcal{A}} \mathbf{B}_1$*,* $x \triangleright^{\mathcal{A}} \mathbf{B}_1 \triangleright^{\mathcal{A}} \mathbf{B}_2$*, ... denote this sequence of random variables drawn from the same experiment with a particular* $\mathcal{A}$*.*

We will be primarily interested in the case of iterating the same beacon $\mathbf{B}$*, i.e., taking* $\mathbf{B}_1 = \cdots = \mathbf{B}_k = \mathbf{B}$*, in which case we write*

$$x \blacktriangleright_k^{\mathcal{A}} \mathbf{B} \;\; \textit{as a shorthand for} \;\; x \underbrace{\triangleright^{\mathcal{A}} \mathbf{B} \triangleright^{\mathcal{A}} \cdots \triangleright^{\mathcal{A}} \mathbf{B}}_{k},$$

with the convention that $x \blacktriangleright_0^{\mathcal{A}} \mathbf{B} = x$*; as above, we also let* $x \blacktriangleright_k^{\mathcal{A}} \mathbf{B}$ *refer to the respective random experiment. We omit the superscripts* $\mathcal{A}$ *whenever the adversary is clear from the context.*

3 Generic Min-Entropy Bound for Iterative Beacons

We are now ready to state and prove our generic beacon composition theorem.

Theorem 1 (Beacon Composition). *Let $\mathbf{B} = (\mathcal{G}_x, \epsilon_x)_{x\in\{0,1\}^\lambda}$ be a beacon with security parameter λ and natural failure rate ϵ. Let $\mathcal{A}$ be any (information-theoretic) adversary. Assume that*

$$\Pr[\mathcal{G}(\mathbf{B}) > \Gamma] \leq \gamma \tag{4}$$

for some $\Gamma \in \mathbb{N}$ and $\gamma \in [0,1]$. Let X_0 be uniformly distributed over $\{0,1\}^\lambda$ and let $X_i = X_0 \blacktriangleright_i^{\mathcal{A}} \mathbf{B}$ for $i \in [k]$ be the output of the i-th invocation of $\mathbf{B}$ in the random experiment $X_0 \blacktriangleright_k^{\mathcal{A}} \mathbf{B}$ involving $\mathcal{A}$. Then there is an event E with probability no more than $k\Gamma(\gamma+\epsilon)$ in the random coins of $X_0 \blacktriangleright_k^{\mathcal{A}} \mathbf{B}$ for which

$$H_\infty\big(X_k \mid \overline{E}\big) \geq \lambda - \log \Gamma\,.$$

Moreover, this guarantee persists under conditioning by prior beacon values:

$$H_\infty\big(X_k \mid X_0, \ldots, X_{k-1}, \overline{E}\big) \geq \lambda - \log \Gamma\,.$$

In particular, the random variable X_k has $2k\Gamma(\gamma+\epsilon)$-smooth min entropy of at least $\lambda - \log \Gamma$.

Proof. Let the random variables X_i be as described in the statement of the theorem and let G_i be the associated random variables determined in Step (G1) of the i-th invocation of the grinding game during $X_0 \blacktriangleright_k^{\mathcal{A}} \mathbf{B}$.

We proceed by induction on k. For $k = 1$, let $E_\perp$ and E_{tail} denote the bad events that $X_0 \triangleright^{\mathcal{A}} \mathbf{B} = \perp$ and $\mathcal{G}_{X_0} > \Gamma$, respectively. Since X_0 is uniform in $\{0,1\}^\lambda$, we have $\Pr[E_\perp] = \epsilon(\mathbf{B}) = \epsilon$ by Definition 3 and $\Pr[E_{\text{tail}}] < \gamma$ by Equation (4). Recall that $G_1 = \mathcal{G}_{X_0}$. Define $E_1 = E_\perp \vee E_{\text{tail}}$ and note that $\Pr[E_1] \leq (\gamma+\epsilon) \leq (\gamma+\epsilon)\Gamma$. Conditioning on $\overline{E_1} = \overline{E_\perp} \wedge \overline{E_{\text{tail}}}$, the random variable $X_1 = X_0 \triangleright \mathbf{B}$ is G_1-optative, and hence Γ-optative, by construction; we conclude that $H_\infty(X_1 \mid \overline{E_1}) \geq \lambda - \log(\Gamma)$ by Lemma 1(iii) as desired. Observe that the internal random variables U_i defining the result in the grinding game associated with X_k (when the result is G_1-optative) are independent of X_0. The amplified version of the conclusion then follows: $H_\infty(X_1 \mid X_0, \overline{E_1}) \geq \lambda - \log(\Gamma)$.

Turning to the induction case, we prove a strengthened form that explicitly identifies the event E_k under which the min-entropy is bounded by conditioning. For $k \geq 1$, let

$$T_k = \{(x_0, g_1, x_1, \ldots, g_k, x_k) \mid x_i \in (\{0,1\}^\lambda \cup \{\perp\}), g_i \in \mathbb{N}\}$$

be the collection of all possible "transcripts" of $X_0 \blacktriangleright_k^{\mathcal{A}} \mathbf{B}$. In the context of an adversary $\mathcal{A}$ this set is given a probability distribution $\mathcal{B}_k$ by the random variables $X_0, G_1, X_1, \ldots, G_k, X_k$ (determined by the definition of beacon composition). Let

$$E_k = \{(x_0, \ldots, g_k, x_k) \mid \exists i \colon (x_i = \perp \vee g_i > \Gamma)\} \subset T_k\,.$$

(Note that this definition is consistent with the definition of E_1 in the previous paragraph.) We adopt the induction hypothesis that

$$\Pr_{\mathcal{B}_{k-1}}[E_{k-1}] \leq (k-1)\Gamma(\gamma+\epsilon) \tag{5}$$

and

$$H_\infty(X_{k-1} \mid \overline{E_{k-1}}) \geq \lambda - \log \Gamma\,. \tag{6}$$

Observe that, conditioned on $\overline{E_{k-1}}$, the random variable $X_{k-1} \neq \perp$ and hence the "$\perp$-passthrough" clause (G4) in Fig. 1 is not activated for X_k. Then combining the induction assumption (6) with Lemma 2, we conclude that

$$\Pr_{\mathcal{B}_k}[X_k = \perp \mid \overline{E_{k-1}}] \leq \Gamma\epsilon\,. \tag{7}$$

For a value $x \in \{0,1\}^\lambda$, let $A(x) = \Pr_{G \leftarrow \mathcal{G}_x}[G > \Gamma]$ and observe that $\mathbb{E}[A(X)] = \Pr[\mathcal{G}(\mathbf{B}) > \Gamma] \leq \gamma$ where X is drawn according to the uniform distribution in $\{0,1\}^\lambda$. Continuing to condition on $\overline{E_{k-1}}$ and again employing (6), another application of Lemma 2 yields

$$\Pr_{\mathcal{B}_k}[G_k > \Gamma \mid \overline{E_{k-1}}] = \mathbb{E}[A(X_{k-1}) \mid \overline{E_{k-1}}] \leq \gamma\Gamma\,. \tag{8}$$

Combining (5), (7) and (8) along with the definition of E_k then gives us

$$\begin{aligned}\Pr[E_k] &\le \Pr[E_{k-1}] + \Pr[\overline{E_{k-1}}] \cdot \Pr[X_k = \bot \text{ or } G_k > \Gamma \mid \overline{E_{k-1}}] \\ &\le (k-1)\Gamma(\gamma+\epsilon) + \Pr[X_k = \bot \mid \overline{E_{k-1}}] + \Pr[G_k > \Gamma \mid \overline{E_{k-1}}] \\ &\le k\Gamma(\gamma+\epsilon)\,,\end{aligned}$$

where all probabilities are taken over $\mathcal{B}_k$. Conditioning on $\overline{E_k}$, the random variable X_k is G_k-optative (and hence Γ-optative), giving us $H_\infty(X_k \mid \overline{E_k}) \ge \lambda - \log \Gamma$ by Lemma 1(iii) and concluding the proof. As in the base case, the random variables U_i defining the random variable X_k (when it is indeed G-optative) are independent of $X_1, \ldots, X_k$; it follows that $H_\infty(X_k \mid X_0, \ldots, X_{k-1}, \overline{E_k}) \ge \lambda - \log \Gamma$. ◻

Remark 1. The final conclusion can be further strengthened to include conditioning on the entire prior transcript of the beacon protocol, including the values determined by the uniform random variables $(U_1, \ldots, U_g)$ defining the grinding game in the previous iterations. This fact is useful for studying such beacons when they appear as components of other protocols. In particular, examining the definition of sequential composition, define the "full transcript" of $X_0 \blacktriangleright_k^{\mathcal{A}} \mathbf{B}$ to be the data $(x_0, g_1, \vec{u}^{(1)}, x_1, \ldots, g_k, \vec{u}^{(k)}, x_k)$, where each $\vec{u}^{(i)}$ are the sequence of values drawn by the U_i random variables in the definition of the grinding game. Then by the same argument, the proof establishes that $H_\infty(X_k \mid T_{k-1}, \overline{E_k}) \ge \lambda - \log \Gamma$, where T_{k-1} is the full transcript of the first $k-1$ iterations.

Theorem 1 motivates the following asymptotic definition.

Definition 5 (Beacon security under composition). *Let $(\mathbf{B}_{(\lambda)})_{\lambda\in\mathbb{N}}$ be a sequence of grindable beacons so that $\mathbf{B}_{(\lambda)}$ has security parameter λ. The collection $(\mathbf{B}_{(\lambda)})_{\lambda\in\mathbb{N}}$ is* secure under composition *if there is a constant $\delta \in (0,1)$ and a function $\Gamma\colon \mathbb{N} \to \mathbb{N}$ such that:*

1. *the quantity $\Gamma(\lambda) \cdot \left(\Pr\left[\mathcal{G}\left(\mathbf{B}_{(\lambda)}\right) > \Gamma(\lambda)\right] + \epsilon\left(\mathbf{B}_{(\lambda)}\right)\right)$ is negligible in λ, and*
2. *for each $\lambda \in \mathbb{N}$ we have $\log \Gamma(\lambda) \le \delta\lambda$.*

The usefulness of Definition 5 is captured by the following corollary, which is a direct consequence of Theorem 1.

Corollary 1. *Let $\mathbf{B} = (\mathbf{B}_{(\lambda)})_{\lambda\in\mathbb{N}}$ be a sequence of grindable beacons with security parameter λ that is secure under composition (with parameter δ). Consider the random experiment $X_0 \blacktriangleright_k^{\mathcal{A}} \mathbf{B}$ of k-wise sequential composition of $\mathbf{B}$ with any (information-theoretic) adversary $\mathcal{A}$, where $k = \mathsf{poly}(\lambda)$ and X_0 is uniform in $\{0,1\}^\lambda$. Then there exists a sequence of events $(E_\lambda)_{\lambda\in\mathbb{N}}$ such that:*

1. *The probability of E_λ occurring is negligible in λ.*
2. *The output X_k of the beacon's final k-th invocation satisfies $H_\infty\left(X_k \mid \overline{E_\lambda}\right) \ge (1-\delta)\cdot\lambda$.*

4 Warm-Up: Selfish Mixing

We apply our framework to the selfish mixing attack [16]. Our high-level approach is to define a beacon $\mathbf{B}_{\mathsf{sm}}$ that captures this class of grinding attacks, obtain a tail bound of the form (4) on the natural grinding capacity of $\mathbf{B}_{\mathsf{sm}}$, and then use it to prove that for adversaries controlling a minority of stake, $\mathbf{B}_{\mathsf{sm}}$ is secure under composition in the sense of Def. 5.

Description of Selfish Mixing. Consider epochs consisting of $L = \Omega(\lambda)$ slots, and assume that in each slot, there is exactly one *block proposer* that is allowed to produce a block. Each block contains a VRF value produced by the block proposer, and these VRF values appearing in the final chain along the whole epoch are then concatenated and hashed to produce the beacon output for the next epoch. Moreover, we assume *idealized proposer selection*, meaning that under uniform epoch randomness we have that for each slot the probability that a party is selected to be a block proposer for this slot is proportional to its stake.

In the selfish mixing attack, the adversary takes advantage of a tail of the epoch that consists entirely of slots with adversarial block proposers (so-called *adversarial slots*). Assuming last t slots of the epoch are adversarial, the adversary can by the end of slot $L-t$ perform the following local computation: for each subset $S \subseteq \{L-t+1,\ldots,L\}$, he can evaluate what the value of the beacon output would be if the block proposers in slots from S produced their blocks honestly, while the remaining block proposers from slots $\{L-t+1,\ldots,L\} \setminus S$ entirely skipped the block production opportunity and remained silent. This allows the adversary to locally determine 2^t possible beacon outputs, evaluate which one is the most preferable for him, and instruct the adversarially controlled block proposers to act in a way that leads to this preferred output with certainty. In particular, the adversary may choose among the potential beacon outputs based on the length of the tail of adversarial slots that it implies for the subsequent epoch, with the intention to gradually increase his influence by applying the attack repeatedly.

The beacon $\mathbf{B}_{\mathsf{sm}}$. Formally, we capture this behavior by considering a grindable beacon $\mathbf{B}_{\mathsf{sm}}$ which allows for exactly this form of grinding. Towards that, for any input (i.e., epoch randomness) $x \in \{0,1\}^\lambda$, define the *length of the adversarial tail* for x, denoted $T(x)$, to be the number of trailing slots in the epoch that are assigned adversarial block proposers when the epoch randomness governing the assignment is x. (For simplicity of modeling, we assume that x is sufficiently long to information-theoretically determine $T(x)$.) We define $\mathbf{B}_{\mathsf{sm}} = (\mathcal{G}_x^{\mathsf{sm}}, \epsilon_x^{\mathsf{sm}})_{x\in\{0,1\}^\lambda}$ where for each $x \in \{0,1\}^\lambda$, $\mathcal{G}_x^{\mathsf{sm}} = 2^{T(x)}$ and $\epsilon_x^{\mathsf{sm}} \in [0,1]$ may be arbitrary to represent the natural failure rate of the considered protocol allowing for selfish mixing, assuming only that $\epsilon \triangleq \epsilon(\mathbf{B}_{\mathsf{sm}})$ is negligible in λ. Notice that $T(x)$, and hence $\mathbf{B}_{\mathsf{sm}}$, implicitly depends on the adversarial stake share, which we denote α and assume that $\alpha \in [0,1/2)$.

Security of $\mathbf{B}_{\mathsf{sm}}$ under composition. For each $i \in [L]$, let A_i be the indicator random variable of the event that the block proposer in slot i is adversarial under a uniformly random epoch randomness $X \xleftarrow{\$} \{0,1\}^\lambda$. Based on the idealized proposer selection, $\Pr[A_i = 1] = \alpha$ and the variables $\{A_i\}_{i=1}^L$ are independent. We clearly have

$$T(X) = \max\{i \in \{0,\ldots,L\} \mid A_{L-i+1} = A_{L-i+2} = \ldots = A_L = 1\}$$

and hence it is easy to see that for any $t \in \{0,\ldots,L\}$ we have $\Pr[T(X) \geq t] = \alpha^t$. Moreover, any epoch with $T(X) = t$ allows for 2^t grinding attempts via the selfish mixing strategy, i.e., we have

$$\Pr\left[\mathcal{G}(\mathbf{B}_{\mathsf{sm}}) \geq 2^t\right] = \Pr_{X \xleftarrow{\$} \{0,1\}^\lambda}\left[\mathcal{G}_X^{\mathsf{sm}} \geq 2^t\right] = \Pr_{X \xleftarrow{\$} \{0,1\}^\lambda}\left[T(X) \geq t\right] \leq \alpha^t\,, \tag{9}$$

giving us a tail bound on $\mathcal{G}(\mathbf{B}_{\mathsf{sm}})$ of the form (4) for any $t \in \{0,\ldots,L\}$.

To conclude that $\mathbf{B}_{\mathsf{sm}}$ is secure under composition according to Def. 5, and with (9) in mind, let $\mathit{\Gamma}(\lambda) \triangleq 2^t$ for some $t(\lambda) \in \{0,\ldots,L\}$ to be specified later, and fix any $\delta \in (0,1)$. As per Def. 5, it suffices to establish three facts (keeping some dependences on λ implicit):

(i) $\log(\mathit{\Gamma}) \leq \delta\lambda$ for some fixed $\delta \in (0,1)$;
(ii) $\mathit{\Gamma} \cdot \Pr\left[\mathcal{G}\left(\mathbf{B}_{\mathsf{sm}}\right) > \mathit{\Gamma}\right]$ is negligible in λ;
(iii) $\mathit{\Gamma} \cdot \epsilon(\mathbf{B}_{\mathsf{sm}})$ is negligible in λ.

Towards that, define $h(\lambda) := -\log(\epsilon(\lambda))$; since ε is negligible, we have $h(\lambda) = \omega(\log\lambda)$. Pick $t(\lambda) = \min\left\{\lfloor\delta\lambda\rfloor, \lfloor\frac{1}{2}h(\lambda)\rfloor\right\}$. Then we have $\log(\mathit{\Gamma}) = t \leq \delta\lambda$ by definition, satisfying (i). Moreover, since both $\delta\lambda$ and $h(\lambda)$ are $\omega(\log\lambda)$, so is $t(\lambda)$. Based on (9) we have

$$\mathit{\Gamma} \cdot \Pr\left[\mathcal{G}\left(\mathbf{B}_{\mathsf{sm}}\right) \geq \mathit{\Gamma}\right] \leq (2\alpha)^{t(\lambda)} = \exp(t(\lambda)\ln(2\alpha))$$

which is negligible for $\alpha < 1/2$, giving us (ii). Finally,

$$\mathit{\Gamma} \cdot \epsilon(\mathbf{B}_{\mathsf{sm}}) = 2^{t(\lambda)}\epsilon(\lambda) \leq 2^{h(\lambda)/2} \cdot 2^{-h(\lambda)} = 2^{-h(\lambda)/2}\,,$$

satisfying (iii). This establishes the following theorem.

Theorem 2. *If adversarial stake share $\alpha < 1/2$ then the family of beacons $(\mathbf{B}_{\mathsf{sm}})_{\lambda\in\mathbb{N}}$ as defined above is secure under composition.*

Theorem 2 establishes an asymptotic fact implicit in existing concrete analysis of selfish mixing [1], namely that the effects of selfish mixing remain under control as long as the adversarial stake share is sufficiently below 1/2 (see e.g. [1, Fig. 1]). Interestingly, this implies that the use of an iterated randomness beacon susceptible to selfish mixing in consensus protocols that require honest (super)majority in fact does not restrict the security region of the resulting protocol (compared to having access to a perfect beacon).

5 Longest-Chain Rule Beacons and the Case of Cardano

We now apply the framework from Section 3 to proof-of-stake longest-chain rule blockchains and, in particular, the Ouroboros Praos [4] protocol currently powering the Cardano blockchain.[4] While we discuss Ouroboros Praos for concretness and its practical relevance, our analysis in this section fully applies to other longest-chain protocols with similar beacons.

Our main result is Theorem 3 that shows that, for sufficiently weak adversaries (controlling at most 12% of total stake in the case of Cardano), the Praos randomness beacon used iteratively over multiple epochs maintains a negligible probability of failure and—assuming no such failure occurred—a linear (in the security parameter λ) min-entropy of its final output. In particular, the Praos beacon is secure under composition in the sense of Def. 5, we state this as Corollary 2. Moreover, Theorem 4 then extends these conclusions to adversaries controlling up to 15% of stake under the additional assumption that the chain produced by the protocol is sufficiently dense.

5.1 Ouroboros Praos

We start by providing an overview of the protocol sufficient to follow our subsequent arguments.

The protocol, denoted Π_{OP} here, is run by the set of parties $\mathcal{P}$ over a sequence of slots $1, 2, \ldots$ as follows: In each slot s, each party $P \in \mathcal{P}$ evaluates a local, private lottery to determine whether she qualifies as a *slot leader* for the slot s. This is done by evaluating a verifiable random function (VRF [5]) using the secret key associated with P's stake, and providing as inputs to the VRF both the slot index and a so-called *epoch randomness seed* (or epoch seed for short, we will discuss in detail below where this seed comes from). If the VRF output is below a certain threshold proportional to P's stake, then she is an eligible slot leader and can create, sign, and broadcast a block B for that slot, containing new transactions to be included in the ledger. She also includes into the new block the above VRF output and its proof, to certify eligibility to act as a slot leader. Additionally, she also includes in B an additional, independent VRF value ρ_B with a corresponding VRF proof, and looking ahead, the role of these values will be to determine the next epoch seed. Parties participating in the protocol collect all valid blocks and constantly update their states to reflect the longest chain they have seen so far.

Multiple slots are combined into *epochs*, each of which contains $L \in \mathbb{N}$ slots. During epoch j, the slot-leader lottery is based on the stake distribution recorded in the ledger up to the last block of epoch $j-2$. Crucially for our discussion, the lottery is seeded by the *epoch randomness seed* for epoch j, which is derived in a deterministic way from the VRF-values ρ_B that were included into all blocks of the longest chain in the first two thirds of epoch $j-1$. Formally, denoting the epoch randomness seed for epoch j as $\rho^{(j)}$, we have

$$\rho^{(j)} = \mathsf{H}(\rho^{(j-1)} \| j \| \rho_{B_1} \| \rho_{B_2} \| \cdots \| \rho_{B_\ell}) \,,$$

where $B_1, B_2, \ldots, B_\ell$ is the sequence of blocks that appear on the longest chain seen by the end of epoch $j-1$ and correspond to the first $2L/3$ slots of this epoch; and $\mathsf{H}(\cdot)$ denotes a cryptographic hash function.

[4] https://cardano.org

5.2 Characteristic Strings

We describe the outcome of the leader lottery in the above protocol by means of a *characteristic string* [13]. A characteristic string is a string over the alphabet $\Sigma = \mathbb{N} \times \mathbb{N}$; specifically, an execution over L slots gives rise to a characteristic string $w = w_1 \dots w_L \in \Sigma^L$ where, intuitively, each symbol $w_i = (h_i, a_i) \in \Sigma$ indicates that h_i honest parties and a_i adversarial parties were eligible slot leaders for slot i.

For a characteristic string $w = w_1 \dots w_L \in \Sigma^L$ where each $w_i = (h_i, a_i) \in \mathbb{N} \times \mathbb{N}$, we define

$$\#_{\mathsf{h}}(w) \triangleq \sum_{i=1}^{L} h_i \qquad \text{and similarly} \qquad \#_{\mathsf{a}}(w) \triangleq \sum_{i=1}^{L} a_i \,,$$

i.e., the total number of honest and adversarial slot leaders over the sequence of slots corresponding to w. Moreover, we sometimes make use of a similar quantity $\#_{[\mathsf{h}]}(w)$ that denotes the number of symbols in w with positive first coordinate, i.e., $\#_{[\mathsf{h}]}(w) \triangleq |\{i \in [L] \mid w_i \notin \{0\} \times \mathbb{N}\}|$, intuitively, this is the number of slots with at least one honest leader.

Informally, we say that an execution of the protocol Π_{OP} is *consistent* with a characteristic string w if the outcome of the leadership lottery in that execution is captured by w.

5.3 Distribution of the Leader Lottery Outcome

We now describe the distribution of the characteristic string induced by the lottery mechanism when seeded with a uniformly random epoch seed. Let $\alpha \in (0, 1/2)$ and assume that the adversarial players control an α-fraction of the total stake.

We assume that the slot leader-selection lottery (under a uniformly random epoch seed $X \xleftarrow{\$} \{0,1\}^\lambda$) is an ideal private sortition, where the probability that any party becomes a leader in a slot is proportional to its stake, and this probability is independent of any other parties or any other slot.[5] (Note that this leads to slots with no, or several, slot leaders.) Thus, the total number of slot leaders in a particular slot is given by a sum of Bernoulli random variables, one for each party. For sufficiently many parties, this sum of Bernoulli random variables can be approximated by a Poisson-distributed random variable. More concretely, we model the numbers of honest (resp. adversarial) leaders in a given slot as Poisson random variables with parameter r_{h} (resp. r_{a}), where r_{h} and r_{a} satisfy

$$\alpha = \frac{r_{\mathsf{a}}}{r_{\mathsf{h}} + r_{\mathsf{a}}}$$

and $\frac{1}{r_{\mathsf{h}}+r_{\mathsf{a}}}$ is the expected inter-block time in slots (equal to 20 in Cardano). Given the parameters r_{h} and r_{a}, we define the distribution $\mathcal{L}_L(r_{\mathsf{h}}, r_{\mathsf{a}})$ of length-L characteristic strings $w = w_1 \dots w_L \in \Sigma^L$ as follows.

Definition 6 (Distribution $\mathcal{L}_L(r_{\mathsf{h}}, r_{\mathsf{a}})$). *For $L \in \mathbb{N}$, $\mathcal{L}_L(r_{\mathsf{h}}, r_{\mathsf{a}})$ is a distribution on characteristic strings $w = w_1 \dots w_L$ so that for each $i \in [L]$ we have $w_i = (h_i, a_i) \in \mathbb{N} \times \mathbb{N}$, where each h_i (resp. a_i) is an independent Poisson-distributed random variable with parameter r_{h} (resp. r_{a}).*

For simplicity of notation, below we often employ the quantity

$$p_{\mathsf{h}} \triangleq 1 - e^{-r_{\mathsf{h}}}$$

that corresponds to the probability that a fixed slot contains at least one honest success. In fact, our main technical statements will contain a condition relating r_{a} to p_{h} (rather than to r_{h}), as this most naturally dovetails with our analysis. For practical interpretability of our results, these conditions translate to a direct relation on $r_{\mathsf{a}}, r_{\mathsf{h}}$ (i.e., a standard honest majority-type assumption) as described in Table 1, assuming the additional constraint $r_{\mathsf{h}} + r_{\mathsf{a}} = 1/20$ used in Cardano.

[5] Notice that to achieve this information-theoretic guarantee, one needs to assume a sufficient seed length λ to provide enough randomness to deterministically prescribe the characteristic string. We make this assumption to avoid the need to consider a pseudorandom expansion of the seed.

Condition on $p_{\mathsf{h}}, r_{\mathsf{a}}$	Condition on $\alpha \triangleq r_{\mathsf{a}}/(r_{\mathsf{h}} + r_{\mathsf{a}})$
$p_{\mathsf{h}} > 18.6r_{\mathsf{a}}$	$\alpha < 4.98\%$
$p_{\mathsf{h}} > 4er_{\mathsf{a}}$	$\alpha < 8.24\%$
$p_{\mathsf{h}} > 6.91r_{\mathsf{a}}$	$\alpha < 12.40\%$
$p_{\mathsf{h}} > 2er_{\mathsf{a}}$	$\alpha < 15.25\%$

Table 1. Various conditions on $p_{\mathsf{h}}, r_{\mathsf{a}}$ appearing in the analysis, and the corresponding sufficient conditions on the adversarial stake ratio α under the Cardano parametrization $r_{\mathsf{h}} + r_{\mathsf{a}} = 1/20$.

5.4 The Grindable Beacon of Praos

Given the operational description of the randomness beacon built into the Ouroboros Praos protocol from Section 5.1, we now cast it as a grindable beacon

$$\mathbf{B}_{\mathsf{OP}} = (\mathcal{G}_x^{\mathsf{OP}}, \epsilon_x^{\mathsf{OP}})_{x \in \{0,1\}^\lambda}$$

in the sense of Def. 2.

Intuitively, the input $x \in \{0,1\}^\lambda$ to the beacon $\mathbf{B}_{\mathsf{OP}}$ corresponds to the epoch randomness seed produced by the beacon execution in the previous epoch (say $i-1$) of the protocol. A fixed seed x determines the outcome of the leadership lottery for the present epoch i, and hence also the characteristic string w_x describing it. Therefore, roughly speaking, the value ϵ_x^{OP} describes whether a failure of the protocol might occur in an epoch with lottery outcome described by the characteristic string w_x, whereas the distribution $\mathcal{G}_x^{\mathsf{OP}}$ captures the number of beacon outputs that the adversary might choose from in an execution consistent with w_x. We now define both ϵ_x^{OP} and $\mathcal{G}_x^{\mathsf{OP}}$ formally.

Defining ϵ_x^{OP}. Given a fixed characteristic string w_x, the original analysis of Praos [4] determines whether there exists any execution of the protocol consistent with w_x that would lead to violation of any of the necessary blockchain properties (common prefix, chain growth, chain quality [8]) resulting into a breakdown of the ledger's safety or liveness, and by extension also a failure of the beacon in the current epoch i.

To capture this formally, we introduce an event F that corresponds to such a failure of a single-epoch execution of Praos. Based on [4], we define F as

$$F = F_{\mathsf{CG}} \vee F_{\mathsf{CP}} \vee F_{\mathsf{CQ}} \tag{10}$$

for the following three sub-events (with an implicit fixed parameter $\tau \in (0,1)$):

- F_{CG} (chain growth violation): Some honest party at some point during the execution holds a chain C as its longest chain, and C contains less than $\tau L/3$ blocks blocks over some sequence of $L/3$ slots.
- F_{CP} (common-prefix violation): There exist slots $s_1, s_2 \in \mathbb{N}$, honest parties $P_1, P_2 \in \mathcal{P}$ and chains C_1, C_2 such that each P_i holds C_i as its longest chain in slot s_i, and neither of $C_1^{\lfloor \tau L/3} \prec C_2$, $C_2^{\lfloor \tau L/3} \prec C_1$ is satisfied, where $C^{\lfloor d}$ represents the chain C after removing its suffix consisting of blocks from the most recent d slots.
- F_{CQ} (chain quality violation): Some honest party at some point during the execution holds a chain C as its longest chain, and C contains no honestly created blocks over some sequence of $L/3$ slots.

See [2, Theorem 7] for an exposition of how the absence of these failure events implies a secure iterative execution of Π_{OP} and its beacon. Based on that, we define ϵ_x^{OP} as

$$\epsilon_x^{\mathsf{OP}} = \begin{cases} 1 & \text{if there exists a single-epoch execution consistent with } w_x \text{ in which} \\ & F \text{ occurs,} \\ 0 & \text{otherwise.} \end{cases}$$

Defining $\mathcal{G}_x^{\mathsf{OP}}$. As noted above, $\mathcal{G}_x^{\mathsf{OP}}$ is the number of beacon outcomes that the adversary may choose from at the end of a single-epoch execution seeded with x, and hence consistent with w_x. (Note that if $\epsilon_x^{\mathsf{OP}} = 1$ then $\mathcal{G}_x^{\mathsf{OP}}$ may take an arbitrary fixed value as it does not affect the outcome of the game $\mathsf{GG}^{\mathbf{B}_{\mathsf{OP}},\mathcal{A}}(x)$.) To capture this formally, we need to define so-called *option sequences*:

Definition 7 (Option sequences). *Let $w = w_1 \dots w_n \in \Sigma^n$ be a characteristic string of length n. An* option sequence *for w is a tuple of pairs $\{(i_j, b_j)\}_{j=0}^{\ell}$ satisfying the following conditions:*

(os1) $w_{i_0} = (h_{i_0}, a_{i_0})$ *satisfies* $b_0 \in [h_{i_0}]$,
(os2) $1 \leq i_0 < i_1 < \cdots < i_\ell \leq n$,
(os3) *for each* $j \in [\ell]$ *the symbol* $w_{i_j} = (h_{i_j}, a_{i_j})$ *satisfies* $b_j \in [a_{i_j}]$.
(os4) $\ell \geq \#_{[\mathsf{h}]}(w_{i_0+1} \dots w_n)$.

We define $\mathsf{os}(w)$ *to be the number of distinct option sequences for* w.

To intuitively appreciate the definition of an option sequence, consider $w = w_1 \dots w_{2L/3} \in \Sigma^{2L/3}$ to be the characteristic string describing the lottery outcome during the first $2L/3$ slots of an epoch, i.e., exactly the region from which the VRF values are collected for the beacon computation. In that setting, each option sequence $\{(i_j, b_j)\}_{j=0}^{\ell}$ for w directly corresponds to a sequence of $\ell+1$ blocks in this execution that is at the same time:

- starting with an honest block from slot i_0 of w corresponding to the b_0-th out of the h_{i_0} honest successes from that slot (os1);
- continuing with adversarial blocks over an increasing sequence of slots (os2), each block corresponding to the b_j-th out of the a_{i_j} adversarial successes from the respective slot (os3); and finally
- is length-competitive compared to the honest chain (os4).

It is easy to see that such a sequence can—by producing the respective adversarial blocks so that they form a chain—result exactly in a private fork giving the adversary one *option* to choose from when grinding, as publishing this chain would result in the beacon taking the value determined by the VRF values contained in that chain.

However, notice that this does not necessarily cover all the grinding options the adversary might be choosing from. Namely, even chains that satisfy conditions (os1)–(os3) but are slightly shorter than the honest chain might still become competitive after the cut-off slot $s = 2L/3$ if, say, the adversary is lucky to have a few lottery victories immediately after the slot s: she can use those to extend the chain in question to become length-competitive.

Fortunately, the accounting can be easily extended to cover this phenomenon. For the above to occur, it would have to be the case that there exists a slot $s' > s$ and from the perspective of the execution up to s', the chain in question (or rather, its adversarial extension) has become competitive. Hence, if we now let $w \in \Sigma^L$ denote the characteristic string describing the lottery outcomes for the whole epoch, then summing option sequences for all prefixes of w of length at least $2L/3$ will clearly capture this scenario as well. Notice that this approach leads to over-counting grinding opportunities (as we also count options that differ only after slot s and hence lead to the same beacon output), but this is acceptable as we are aiming to upper-bound grinding.

Therefore, we define

$$\mathcal{G}_x^{\mathsf{OP}} = \sum_{i=2L/3}^{L} \mathsf{os}\left((w_x)_{1:i}\right) , \tag{11}$$

where $(w_x)_{1:i}$ denotes the length-i prefix of w_x.

Looking ahead, we are interested in applying Theorem 1 to $\mathbf{B}_{\mathsf{OP}}$, and hence we need to understand both its natural failure rate $\mathcal{G}(\mathbf{B}_{\mathsf{OP}})$, as well as its natural grinding capacity $\epsilon(\mathbf{B}_{\mathsf{OP}})$. We tackle these two questions in Sections 5.5–5.8.

5.5 Natural Failure Rate of $\mathbf{B}_{\mathsf{OP}}$

The natural failure rate $\epsilon(\mathbf{B}_{\mathsf{OP}})$ corresponds to the probability that a single-epoch execution of the Praos protocol, when seeded with a uniformly random epoch randomness seed, leads to the failure event F occurring. This probability has been studied in detail in [4] and subsequent works, which we can invoke for the following statement (see also [2, Theorem 7]).

Proposition 1 (Security of Single-Epoch Praos [4]). *Let $\alpha \in [0, 1/2)$. Consider a single-epoch execution of the Ouroboros Praos protocol with uniform epoch seed, in the presence of an adversary controlling an α-fraction of stake and unable to break the cryptographic building blocks of the protocol. Then we have*

$$\epsilon(\mathbf{B}_{\mathsf{OP}}) = \Pr[F] \leq \exp(-\Theta(\lambda)) .$$

5.6 Natural Grinding Capacity of $\mathbf{B}_{\mathsf{OP}}$

Obtaining a tail bound for the natural grinding capacity $\mathcal{G}(\mathbf{B}_{\mathsf{OP}})$ will be our main technical task, as this quantity has not been studied by any previous analysis of the protocol. Recall that $\mathcal{G}(\mathbf{B}_{\mathsf{OP}})$ is the quantity $\mathcal{G}_X$ for a uniformly random epoch seed X, i.e., based on (11) we have

$$\mathcal{G}(\mathbf{B}_{\mathsf{OP}}) = \sum_{i=2L/3}^{L} \mathsf{os}\left(W_{1:i}\right) , \tag{12}$$

where $W \in \Sigma^L$ is a random variable describing the whole epoch's lottery outcome under a uniform seed, i.e., $W \sim \mathcal{L}_L(r_{\mathsf{h}}, r_{\mathsf{a}})$, and $W_{1:i}$ is again its length-i prefix. We proceed by deriving a tail bound for $\mathsf{os}(W_{1:i})$ for an arbitrary i, and account for the effect of the summation (12) later. For simplicity of presentation, we will describe the case for $\mathsf{os}(W)$, but our treatment will apply to any of the prefixes of W appearing in (12).

5.7 Moments of the Random Variable $\mathsf{os}(W)$

To obtain a tail bound on the quantity $\mathsf{os}(W)$, we will use the following fact from probability theory.

Fact 1 *For any non-negative random variable X and any $\mu > 0$, we have $\Pr[X \geq t] \leq \mathbb{E}[X^\mu]/t^\mu$.*

To apply Fact 1, we first need to express the μ-th moment $\mathbb{E}[\mathsf{os}(W)^\mu]$ of $\mathsf{os}(W)$. We will achieve this in Equation (16), and then we set out to upper-bound this quantity in Section 5.8, which will be the heart of our technical contribution in this section.

Let $y = y_1 \dots y_n \in \Sigma^n$ be a characteristic string and denote the components of each y_j as $y_j = (h_j, a_j)$. Define

$$S(y) \triangleq \sum_{\substack{J \subseteq [|y|] \\ |J| \geq \#_{[\mathsf{h}]}(y)}} \prod_{j \in J} a_j , \tag{13}$$

where the sum goes over all subsets of $1, 2, \dots, |y|$ of size at least $\#_{[\mathsf{h}]}(y)$.

Notice that if we consider above a characteristic string y that is itself a suffix of another string $w = xy \in \Sigma^L$ for $L \geq n$, then $S(y)$ is the number of ways that one can extend any chain appearing in x by an adversarial-only suffix over y, so that the resulting chain is "length-competitive". Namely, the chain over x can be extended by adding a path of adversarial-only blocks associated with slots from J in an increasing order, and the length-competitiveness will be guaranteed by the condition $|J| \geq \#_{[\mathsf{h}]}(y)$. Note that if an adversary controls a_j leaders in a given slot j, she can choose which of them produces the block for this slot (leading to different VRF values in the block), which is why we multiply the values a_j over all $j \in J$.

By summing over all honest slots, the total number of option sequences for $w = w_1 w_2 \cdots w_n$ is hence

$$\mathsf{os}(w) = \sum_{i=1}^{L} h_i \cdot S(w_{i+1} \cdots w_L)\,. \tag{14}$$

(Notice the change in indexing: h_i is now the first component of the i-th symbol $w_i = (h_i, a_i)$ of the whole string w.)

We introduce the helper notation

$$T(b,c) \triangleq \sum_{i=b}^{c} \binom{c}{i}$$

and observe that

$$S(y) \leq \sum_{i=\#_{[\mathsf{h}]}(y)}^{\#_{\mathsf{a}}(y)} \binom{\#_{\mathsf{a}}(y)}{i} = T\left(\#_{[\mathsf{h}]}(y), \#_{\mathsf{a}}(y)\right)\,. \tag{15}$$

Recall that W denotes the random variable distributed according to $\mathcal{L}_L(r_{\mathsf{h}}, r_{\mathsf{a}})$. We will refer to its symbols as $W = W_1 \ldots W_L$ with $W_i = (H_i, A_i)$ for $i \in [L]$ and let $H_0 = 1$ by convention; we also use the notation $W_{i:j} \triangleq W_i \ldots W_j$. Given (14), we can express the μ-th moment of $\mathsf{os}(W)$ as

$$\begin{aligned}
\mathbb{E}[\mathsf{os}(W)^\mu] &= \mathbb{E}\left[\left(\sum_{i=1}^{L} H_i \cdot S(W_{i+1:L})\right)^{\mu}\right] \\
&\overset{(a)}{\leq} \left(\sum_{i=1}^{L}\Big(\mathbb{E}\left[(H_i \cdot S(W_{i+1:L}))^\mu\right]\Big)^{\frac{1}{\mu}}\right)^{\mu} \overset{(b)}{\leq} L^{\mu-1} \cdot \sum_{i=1}^{L} \mathbb{E}\left((H_i \cdot S(W_{i+1:L}))^\mu\right) \\
&\overset{(c)}{\leq} L^{\mu-1} \cdot \sum_{i=1}^{L} \sum_{k\geq 1} \sum_{b=0}^{L-i} \sum_{c \geq b} \Pr[H_i = k] \cdot \Pr[\#_{[\mathsf{h}]}(W_{i+1:L}) = b] \cdot \Pr[\#_{\mathsf{a}}(W_{i+1:L}) = c] \cdot (k \cdot T(b,c))^\mu\,,
\end{aligned}$$

where inequality (a) uses Minkowski's inequality; inequality (b) leverages the fact that for any $a_i \geq 0$ and $\mu \geq 1$ we have

$$\left(\sum_{i=1}^{L} a_i^{1/\mu}\right)^{\mu} \leq L^{\mu-1} \sum_{i=1}^{L} a_i\,,$$

which follows from Jensen's inequality applied to the convex function $\varphi(x) = x^\mu$; and inequality (c) relies on (15) and the independence of the random variables $\{H_i, A_i\}_{i\in[L]}$. Let

$$\begin{aligned}
A(\mu) &\triangleq \sum_{k\geq 1} \Pr[H_1 = k] \cdot k^\mu = \mathbb{E}[H_1^\mu]\,, \\
B_\ell(b) &\triangleq \Pr_{V \leftarrow \mathcal{L}_\ell(r_{\mathsf{h}}, r_{\mathsf{a}})}\left[\#_{[\mathsf{h}]}(V) = b\right]\,, \\
C_\ell(c) &\triangleq \Pr_{V \leftarrow \mathcal{L}_\ell(r_{\mathsf{h}}, r_{\mathsf{a}})}\left[\#_{\mathsf{a}}(V) = c\right]\,,
\end{aligned}$$

then we have

$$\mathbb{E}[\mathsf{os}(W)^\mu] \leq L^{\mu-1} \cdot A(\mu) \cdot \sum_{t=0}^{L} \sum_{b=0}^{t} \sum_{c\geq b} B_t(b) \cdot C_t(c) \cdot T(b,c)^\mu\,. \tag{16}$$

5.8 Bounding the μ-th Moment

In this section we consider a value $\mu \in [1,2]$ and focus on upper-bounding the μ-th moment $\mathbb{E}[\mathsf{os}(W)^\mu]$ based on (16).

Lemma 3. *For $\mu \in [1,2]$ we have $A(\mu) \le (r_{\mathsf{h}} + r_{\mathsf{h}}^2)^{\mu/2}$.*

Proof. Recall that $A(\mu) = \mathbb{E}[H_1^\mu]$ and H_1 is a Poisson-distributed random variable with parameter r_{h}, hence $\mathbb{E}[H_1] = \mathbb{V}[H_1] = r_{\mathsf{h}}$ and $\mathbb{E}[H_1^2] = \mathbb{V}[H_1] + \mathbb{E}[H_1]^2 = r_{\mathsf{h}} + r_{\mathsf{h}}^2$. Finally, as $\mu/2 \in [1/2, 1]$, the function $x \mapsto x^{\mu/2}$ is concave, and hence Jensen's inequality gives us $\mathbb{E}[H_1^\mu] = \mathbb{E}\big[(H_1^2)^{\frac{\mu}{2}}\big] \le \big(\mathbb{E}[H_1^2]\big)^{\mu/2} = \big(r_{\mathsf{h}} + r_{\mathsf{h}}^2\big)^{\mu/2}$ as desired. ◻

The following lemma is upper-bounding an individual summand in (16) and is the central technical step in obtaining Theorem 3. For interpreting the "honest-majority" condition $p_{\mathsf{h}} \ge (11.69\mu - 4.78) \cdot r_{\mathsf{a}}$, recall that $p_{\mathsf{h}} \triangleq 1 - e^{-r_{\mathsf{h}}}$ and Table 1.

Lemma 4. *Let $\mu \in [1,2]$ and assume that $p_{\mathsf{h}} \ge (11.69\mu - 4.78) \cdot r_{\mathsf{a}}$. Then there exists a constant $R > 0$ depending on $\mu, p_{\mathsf{h}}, r_{\mathsf{a}}$ such that for any $t \in \mathbb{N}$, $b \in \{0, \dots, t\}$ and $c \ge b$ we have*

$$B_t(b) \cdot C_t(c) \cdot T(b,c)^\mu \le \exp(-Rt)\,.$$

Proof. Consider the quantity

$$P_t(b,c) \triangleq B_t(b) \cdot C_t(c) \cdot (T(b,c))^\mu = \binom{t}{b} p_{\mathsf{h}}^b (1-p_{\mathsf{h}})^{t-b} e^{-tr_{\mathsf{a}}} \frac{(tr_{\mathsf{a}})^c}{c!} \left(\sum_{i=b}^{c} \binom{c}{i} \right)^\mu .$$

To simplify notation, let us introduce variables β and γ such that $b = \beta t$, $c = \gamma t$, and note that $0 \le \beta \le 1$ and $\gamma \ge \beta$. We now upper-bound each of the individual factors in $P_t(b,c)$ and focus on their exponents: [6]

$$\begin{aligned}
B_t(b) &= \binom{t}{b} p_{\mathsf{h}}^b (1-p_{\mathsf{h}})^{t-b} \le \exp\left(H(\beta)\ln(2)t\right) \cdot \exp(\beta \ln(p_{\mathsf{h}})t) \cdot \exp((1-\beta)\ln(1-p_{\mathsf{h}})t) \\
&= \exp\left[\left(\beta \ln \frac{p_{\mathsf{h}}}{\beta} + (1-\beta) \ln \frac{1-p_{\mathsf{h}}}{1-\beta} \right) \cdot t \right], \\
C_t(c) &= e^{-tr_{\mathsf{a}}} \frac{(tr_{\mathsf{a}})^c}{c!} \le \exp\left[\left(\gamma - r_{\mathsf{a}} - \gamma \ln \frac{\gamma}{r_{\mathsf{a}}} \right) \cdot t \right], \\
(T(b,c))^\mu &\le (2^{\gamma t})^\mu = \exp(\ln(2)\mu\gamma t)\,,
\end{aligned}$$

where the upper bound on $C_t(c)$ follows by using $n! \ge (n/e)^n$. Putting these estimates together, we get

$$P_t(b,c) \le \exp\left(\beta \ln \frac{p_{\mathsf{h}}}{\beta} + (1-\beta) \ln \frac{1-p_{\mathsf{h}}}{1-\beta} - \gamma \ln \frac{\gamma}{r_{\mathsf{a}}} + (1 + \mu \ln(2))\gamma - r_{\mathsf{a}} \right) \cdot t\,.$$

Considering the term from the exponent, define

$$Q(\beta,\gamma) \triangleq \beta \ln \frac{p_{\mathsf{h}}}{\beta} + (1-\beta) \ln \frac{1-p_{\mathsf{h}}}{1-\beta} - \gamma \ln \frac{\gamma}{r_{\mathsf{a}}} + (1 + \mu \ln(2))\gamma - r_{\mathsf{a}}\,, \tag{17}$$

our goal will be to show that $Q(\beta,\gamma)$ is upper-bounded by a negative constant.

First, notice that the derivative of $Q(\beta,\gamma)$ with respect to γ is

$$\frac{dQ(\beta,\gamma)}{d\gamma} = \ln\left(\frac{2\mu r_a}{\gamma}\right) \tag{18}$$

[6] Here and below, for convenience we interpret all entropy-style terms at the boundary by continuity (with $0 \ln 0 := 0$). In particular, for $\beta \in \{0,1\}$ we understand

$$\beta \ln \frac{p_h}{\beta} + (1-\beta) \ln \frac{1-p_h}{1-\beta} = \begin{cases} \ln(1-p_h), & \beta = 0, \\ \ln(p_h), & \beta = 1. \end{cases}$$

which satisfies $dQ(\beta,\gamma)/d\gamma = 0$ at $\gamma = 2\mu r_{\mathsf{a}}$, and one can verify that the second derivative $d^2Q(\beta,\gamma)/d^2\gamma$ is negative at this point, hence $Q(\beta,\cdot)$ has a maximum at $\gamma = 2\mu r_{\mathsf{a}}$, in which case

$$-\gamma \ln\left(\frac{\gamma}{r_{\mathsf{a}}}\right) + (1+\mu\ln 2)\gamma = \left[2\mu \ln\left(\frac{e2^{\mu}}{2\mu}\right)\right] \cdot r_{\mathsf{a}} \leq 2\mu r_{\mathsf{a}}\,.$$

This establishes the upper bound

$$Q(\beta,\gamma) \leq W(\beta) \triangleq \beta \ln\frac{p_{\mathsf{h}}}{\beta} + (1-\beta)\ln\frac{1-p_{\mathsf{h}}}{1-\beta} + (2\mu-1)r_{\mathsf{a}}\,.$$

Recall the constraint $\beta \leq \gamma$. The above argument implies that if $\beta < 2\mu r_{\mathsf{a}}$ then $Q(\beta,\gamma)$ is maximized at $(\beta, 2\mu r_{\mathsf{a}})$, upper-bounded by the value $W(\beta)$ above. Otherwise, $\beta \geq 2\mu r_{\mathsf{a}}$ in which case $Q(\beta,\gamma)$ is maximized at (β,β) taking value

$$L(\beta) \triangleq Q(\beta,\beta) = \beta\ln\frac{p_{\mathsf{h}}}{\beta} + (1-\beta)\ln\frac{1-p_{\mathsf{h}}}{1-\beta} - \beta\ln\frac{\beta}{r_{\mathsf{a}}} + (\mu\ln(2)+1)\beta - r_{\mathsf{a}}\,.$$

Formally,

$$Q(\beta,\gamma) \leq \begin{cases} W(\beta) & \text{if } \beta < 2\mu r_{\mathsf{a}}, \text{ (in fact for all } \beta) \\ L(\beta) & \text{if } \beta \geq 2\mu r_{\mathsf{a}}. \end{cases}$$

We now proceed by handling these two ranges separately; in fact we additionally split the second range into two subcases, resulting in the three cases below.

Case 1: $\beta \in [0, 2\mu r_{\mathsf{a}})$. By assumption we have $p_{\mathsf{h}} > 2\mu r_{\mathsf{a}}$ and we prove that $W(\beta)$ is monotonically increasing for $\beta \in (0, p_{\mathsf{h}})$. We have

$$\frac{\partial W}{\partial \beta}(\beta) = \ln\frac{p_h}{\beta} - \ln\frac{1-p_{\mathsf{h}}}{1-\beta}$$

implying a unique stationary point at $\beta = p_{\mathsf{h}}$. Moreover,

$$\frac{\partial^2 W}{\partial \beta^2}(\beta) = -\frac{1}{1-\beta} - \frac{1}{\beta} < 0$$

for all $\beta \in (0,1)$. We conclude that $\beta = p_{\mathsf{h}}$ is a (unique) maximum in the interval $(0,1)$ and hence that $W(\beta)$ is increasing on the whole interval $(0, p_{\mathsf{h}})$.

Now, define $W(\beta; p_{\mathsf{h}}, r_{\mathsf{a}})$ (merely making the variables explicit in the notation) and consider the parameterization where $p_{\mathsf{h}} = Cr_{\mathsf{a}}$. We are studying the setting where $C > 2\mu$ in which case we have seen that the maximum subject to the condition that $\beta \in (0, 2\mu r_{\mathsf{a}})$ is $W(2\mu r_{\mathsf{a}}; Cr_{\mathsf{a}}, r_{\mathsf{a}})$. Then

$$\frac{\partial W(2\mu r_{\mathsf{a}}, Cr_{\mathsf{a}}, r_{\mathsf{a}})}{\partial r_{\mathsf{a}}}(0) = 2\mu\ln\frac{C}{2\mu} + 4\mu - C - 1$$

and for each $\mu \in [1,2]$ we numerically find the values of C^* for which this term is zero, these are plotted in Figure 2. Note that for $\mu = 1$ we have $C^* \approx 4.7154$ while for $\mu = 2$ we have $C^* \approx 11.0728$. Moreover, for any $C > C^*$ the term is negative.

We then note that

$$\frac{\partial^2 W(2\mu r_{\mathsf{a}}, Cr_{\mathsf{a}}, r_{\mathsf{a}})}{\partial r_{\mathsf{a}}^2}(r_{\mathsf{a}}) = \frac{(C-2\mu)^2}{(2\mu r_{\mathsf{a}}-1)(Cr_{\mathsf{a}}-1)^2} < 0$$

for $r_{\mathsf{a}} < 1/4 \leq 1/2\mu$. It follows that for $C > C^*$, $W(2\mu r_{\mathsf{a}}, Cr_{\mathsf{a}}, r_{\mathsf{a}})$ is maximized over the interval $(0, 1/4)$ at $r_{\mathsf{a}} = 0$. Observe also that

$$\lim_{r_a \to 0} W(2\mu r_{\mathsf{a}}, Cr_{\mathsf{a}}, r_{\mathsf{a}}) = 0\,,$$

hence we can conclude that for any $0 < r_{\mathsf{a}} < 1/4$ and $C > C^*$, $W(2\mu r_{\mathsf{a}}, Cr_{\mathsf{a}}, r_{\mathsf{a}})$ is upper-bounded by a negative constant as desired.

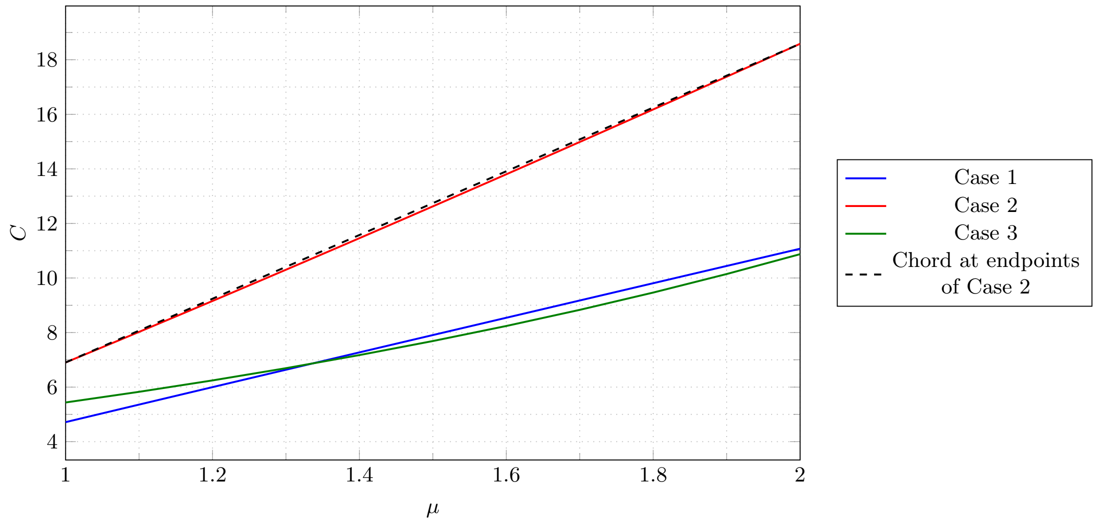

Fig. 2. Lower bounds on $C = p_{\mathsf{h}}/r_{\mathsf{a}}$ arising from Cases 1–3 appearing in the proof of Lemma 4, as a function of μ.

Case 2: $\beta \in [2\mu r_{\mathsf{a}}, p_{\mathsf{h}})$**.** This case forces us to control the function

$$L(\beta) = \beta \ln \frac{p_{\mathsf{h}}}{\beta} + (1-\beta)\ln\frac{1-p_{\mathsf{h}}}{1-\beta} - \beta \ln \frac{\beta}{r_{\mathsf{a}}} + (\mu \ln(2) + 1)\beta - r_{\mathsf{a}}\,.$$

We observe that the first two terms are exactly the quantity bounded by a conventional Chernoff bound and, in particular, when $\beta \le p_{\mathsf{h}}$, we have

$$\beta \ln \frac{p_{\mathsf{h}}}{\beta} + (1-\beta)\ln\frac{1-p_{\mathsf{h}}}{1-\beta} \le -\frac{(\beta - p_{\mathsf{h}})^2}{2p_{\mathsf{h}}}\,.$$

As for the term

$$A(\beta, r_{\mathsf{a}}) \triangleq (\mu \ln(2) + 1)\beta - \beta \ln \frac{\beta}{r_{\mathsf{a}}}\,,$$

we note that by expanding A in a power series in β around the point p_{h}, we have

$$A(\beta, r_{\mathsf{a}}) = \left(p_{\mathsf{h}} - p_{\mathsf{h}} \ln\left(\frac{p_{\mathsf{h}}}{2\mu r_{\mathsf{a}}}\right)\right) - (\beta - p_{\mathsf{h}})\ln\left(\frac{p_{\mathsf{h}}}{2\mu r_{\mathsf{a}}}\right) - \frac{(\beta - p_{\mathsf{h}})^2}{2p_{\mathsf{h}}} + O\left((\beta - p_{\mathsf{h}})^3\right)\,.$$

Note that in the zone $\beta \in (0, p_{\mathsf{h}}]$, this quadratic expansion is an upper bound, hence

$$A(\beta, r_{\mathsf{a}}) \le \left(p_{\mathsf{h}} - p_{\mathsf{h}} \ln\left(\frac{p_{\mathsf{h}}}{2\mu r_{\mathsf{a}}}\right)\right) - (\beta - p_{\mathsf{h}})\ln\left(\frac{p_{\mathsf{h}}}{2\mu r_{\mathsf{a}}}\right) - \frac{(\beta - p_{\mathsf{h}})^2}{2p_{\mathsf{h}}}\,.$$

(Incidentally, this is why we expand around p_{h}, the leftmost point in the range we consider.)

This leads to an upper bound $\hat{L}$ on L (over the range $(r_{\mathsf{a}}, p_{\mathsf{h}})$) of the form

$$\hat{L}(\beta, p_{\mathsf{h}}, r_{\mathsf{a}}) \triangleq 2\beta - \left(\frac{\beta^2}{p_{\mathsf{h}}} + \beta \ln\left(\frac{p_{\mathsf{h}}}{2\mu r_{\mathsf{a}}}\right) + r_{\mathsf{a}}\right)$$

and we find that

$$\frac{\partial \hat{L}}{\partial \beta} = 2 - \frac{2\beta}{p_{\mathsf{h}}} - \ln\left(\frac{p_{\mathsf{h}}}{2\mu r_{\mathsf{a}}}\right) \qquad \text{and} \qquad \frac{\partial^2 \hat{L}}{\partial \beta^2} = -\frac{2}{p_{\mathsf{h}}}\,.$$

As the second derivative is everywhere negative (for $p_{\mathsf{h}} > 0$), it follows that $\hat{L}$ has a unique maximum at the unique point where the first derivative is zero, which occurs at

$$\hat{\beta}^* = p_{\mathsf{h}} \left(1 - \frac{1}{2} \ln\left(\frac{p_{\mathsf{h}}}{2\mu r_{\mathsf{a}}}\right)\right) .$$

We observe that this takes the value zero when $p_{\mathsf{h}} = 2\mu e^2 r_{\mathsf{a}}$ and, moreover, the maximum appears at a negative value of β for $p_{\mathsf{h}} > 2\mu e^2 r_{\mathsf{a}}$. We conclude that for $p_{\mathsf{h}} \geq 2\mu e^2 r_{\mathsf{a}}$, the optimum value of β on the interval $[2\mu r_{\mathsf{a}}, p_{\mathsf{h}}]$ appears at $2\mu r_{\mathsf{a}}$. In this case, we see that

$$\hat{L}(2\mu r_{\mathsf{a}}, p_{\mathsf{h}}, r_{\mathsf{a}}) = r_{\mathsf{a}} \left((4\mu - 1) - \frac{4\mu^2 r_{\mathsf{a}}}{p_{\mathsf{h}}} - 2\mu \ln\left(\frac{p_{\mathsf{h}}}{2\mu r_{\mathsf{a}}}\right)\right)$$

and (continuing to work under the assumption that $p_{\mathsf{h}} \geq 2\mu e^2 r_{\mathsf{a}}$) that

$$\hat{L}(2\mu r_{\mathsf{a}}, p_{\mathsf{h}}, r_{\mathsf{a}}) \leq -\left(1 + \frac{4\mu^2 r_{\mathsf{a}}}{p_{\mathsf{h}}}\right) r_{\mathsf{a}} < 0 .$$

It remains to handle the case with $p_{\mathsf{h}} \in [2\mu r_{\mathsf{a}}, 2\mu e^2 r_{\mathsf{a}})$. We note that the optimum value of $\hat{L}$ at $(\hat{\beta}^*, p_{\mathsf{h}}, r_{\mathsf{a}})$ is

$$\hat{L}^*(p_{\mathsf{h}}, r_{\mathsf{a}}) = \hat{L}(\hat{\beta}^*, p_{\mathsf{h}}, r_{\mathsf{a}}) = p_{\mathsf{h}} - r_{\mathsf{a}} - p_{\mathsf{h}} \ln\left(\frac{p_{\mathsf{h}}}{2\mu r_{\mathsf{a}}}\right) + \frac{1}{4} p_{\mathsf{h}} \left[\ln\left(\frac{p_{\mathsf{h}}}{2\mu r_{\mathsf{a}}}\right)\right]^2$$

and writing $C = p_{\mathsf{h}}/r_{\mathsf{a}}$, we have

$$\hat{L}^*(C r_{\mathsf{a}}, r_{\mathsf{a}}) = r_{\mathsf{a}} \left(-1 + C - C \ln\left(\frac{C}{2\mu}\right) + \frac{C}{4}\left[\ln\left(\frac{C}{2\mu}\right)\right]^2\right) .$$

Evidently, the sign of $\hat{L}^*(C r_{\mathsf{a}}, r_{\mathsf{a}})$ is determined by the sign of

$$\chi(C) = -1 + C - C \ln\left(\frac{C}{2\mu}\right) + \frac{C}{4}\left[\ln\left(\frac{C}{2\mu}\right)\right]^2 = -1 + \frac{C}{4}\left(2 - \ln\left(\frac{C}{2\mu}\right)\right)^2 ,$$

which, for a fixed μ, depends only on $C = p_{\mathsf{h}}/r_{\mathsf{a}}$. We observe that

$$\frac{d\chi}{dC} = -\frac{1}{2} \ln\left(\frac{C}{2\mu}\right) + \frac{1}{4} \ln\left(\frac{C}{2\mu}\right)^2 \qquad \text{and} \qquad \frac{d^2\chi}{dC^2} = \frac{\ln\left(\frac{C}{2\mu}\right) - 1}{2C}$$

and it is easy to see that $d\chi/dC = 0$ at 2μ and $2\mu e^2$ and, additionally, that $C = 2\mu$ is a maximum and $C = 2\mu e^2$ is a minimum of $\chi(\cdot)$. Considering that $\chi(2\mu) = 2\mu - 1 > 0$ and $\chi(2\mu e^2) = -1$, we conclude that there is a unique zero in the range $(2\mu, 2\mu e^2)$. Numerically, this can be determined for each $\mu \in [1, 2]$, and we plot the resulting values in Fig. 2. In particular, we get $C^* = 6.9027...$ for $\mu = 1$ and $C^* = 18.5853...$ for $\mu = 2$.

Finally, to upper-bound $C^*(\mu)$ in every $\mu \in [1, 2]$ by the value of the chord determined by the values of $C^*(\mu)$ in the endpoints $\{1, 2\}$ obtained above, we argue that the function $C^*(\mu)$ assigning to each μ the unique root of $\chi(C, \mu)$ on $(2\mu, 2\mu e^2)$ is convex. Recall the (principal branch of the) Lambert product logarithm function W_0, defined for $z \in [-1/e, \infty)$ to be the unique real $w \geq -1$ for which $we^w = z$. Then

$$\chi(C) = 0 \iff -1 + \frac{C}{4}\Big(\ln\frac{C}{2\mu} - 2\Big)^2 = 0 \iff (L - 2)^2 = \frac{4}{C}$$

where $L := \ln(C/(2\mu))$ and hence $C = 2\mu e^L$. Note from above that $L \in (0, 2)$; writing $y = L - 2$, we have

$$y^2 = \frac{4}{2\mu e^{y+2}} \iff y^2 e^y = \frac{2}{\mu} e^{-2} \implies y e^{y/2} = -\frac{1}{e}\sqrt{\frac{2}{\mu}} \quad (\text{as } y \in (-2, 0)) .$$

Finally, writing $w := y/2$, we obtain

$$we^w = -\frac{1}{e\sqrt{2\mu}} \Rightarrow w = W_0\Big(-\frac{1}{e\sqrt{2\mu}}\Big) \quad (\text{as } w \in (-1,0))\,.$$

We conclude that $C^*(\mu)$, the root in the range $(2\mu, 2\mu e^2)$ can be written as

$$C^* = 2\mu e^L = 2\mu e^2 e^{2w} = 2\mu e^2 \left(\frac{we^w}{w}\right)^2 = \frac{1}{w^2} = \frac{1}{\left[W_0\left(-\frac{1}{e\sqrt{2\mu}}\right)\right]^2}\,.$$

The convexity of $C^*(\mu)$ now follows from known properties of W_0 and standard composition rules. This allows us to upper-bound $C^*(\mu)$ in every $\mu \in [1,2]$ by the value of the chord $11.69\mu - 4.98$ determined by the values of $C^*(\mu)$ in the endpoints $\{1,2\}$ obtained above (cf. Fig. 2).

Case 3: $\beta \geq p_{\mathsf{h}}$**.** To handle the regime where $\beta > p_{\mathsf{h}}$, we note that in this case the binomial part of the bound is irrelevant, since we are above the expected value of the number of honest slots. (This is the part of the bound handled by the Chernoff bound above.) It remains to understand the function

$$A(\beta, r_{\mathsf{a}}) = (\mu \ln(2) + 1)\beta - \beta \ln \frac{\beta}{r_{\mathsf{a}}}\,.$$

We remark that

$$\frac{\partial A}{\partial \beta} = \mu \ln(2) - \ln(\beta/r_{\mathsf{a}}) = \ln\left(2^\mu r_{\mathsf{a}}/\beta\right)$$

and observe that for $\beta > 2^\mu r_{\mathsf{a}}$, this is negative. It follows that there can be only one zero of $A(\cdot, r_{\mathsf{a}})$ in the interval $(2^\mu r_{\mathsf{a}}, \infty)$. By inspection, we see that $A(2^\mu e r_{\mathsf{a}}, r_{\mathsf{a}}) = 0$ and conclude that $A(\beta, r_{\mathsf{a}}) \leq 0$ for $\beta \geq 2^\mu e r_{\mathsf{a}}$. Returning to the development above, if $p_{\mathsf{h}} \geq 2^\mu e r_{\mathsf{a}}$ then $L(\beta, p_{\mathsf{h}}, r_{\mathsf{a}}) \leq 0$ for all $\beta \geq p_{\mathsf{h}}$, as desired. Note that $2^\mu e \approx 5.4366$ for $\mu = 1$ and $2^\mu e \approx 10.8731$ for $\mu = 2$, we plot all values for $\mu \in [1,2]$ in Fig. 2. ◻

To address the unbounded range of the inner-most sum in (16), we prove the following lemma.

Lemma 5. *Fix $\mu \in [1,2]$. For any $b \in \{0, \ldots, t\}$ and any $\delta \geq 2\mu e r_{\mathsf{a}}$ we have*

$$\sum_{c \geq \delta t} B_t(b) \cdot C_t(c) \cdot T(b,c)^\mu \leq \exp\left(Q(\beta,\delta) \cdot t\right)$$

for $Q(.,.)$ as defined in (17).

Proof. Recalling from (18) that $\partial Q(\beta,\gamma)/\partial\gamma = \ln(2\mu r_{\mathsf{a}}/\gamma)$, we observe that for $\gamma \geq 2\mu e r_{\mathsf{a}}$, $\partial Q(\beta,\gamma)/\partial\gamma \leq -1$. It follows that for $\beta > 0$ and $\gamma \geq \delta \geq 2\mu e r_{\mathsf{a}}$ we have $Q(\beta,\gamma) \leq Q(\beta,\delta) - (\gamma - \delta)$. Therefore, recalling $c = \gamma t$, we get

$$\sum_{c \geq \delta t} P_t(b,c) \leq \int_\delta^\infty e^{Q(\beta,\gamma)\cdot t}\, d\gamma \leq \int_\delta^\infty e^{Q(\beta,\delta)\cdot t - (\gamma-\delta)t}\, d\gamma = e^{Q(\beta,\delta)\cdot t} \int_\delta^\infty e^{-(\gamma-\delta)t}\, d\gamma = e^{Q(\beta,\delta)\cdot t} \cdot \frac{1}{t} \leq e^{Q(\beta,\delta)\cdot t}$$

as desired. ◻

To upper-bound (16), let us first inspect the sum $\sum_{c \geq b} P_t(b,c)$ and split the argument into two cases, depending on whether $b < 2\mu e r_{\mathsf{a}} t$ (that is, whether $\beta < 2\mu e r_{\mathsf{a}}$).

If that is the case then applying Lemmas 4 and 5 gives us

$$\begin{aligned}\sum_{c \geq b} P_t(b,c) &\leq \sum_{c=b}^{2\mu e r_{\mathsf{a}} t} P_t(b,c) + \sum_{c \geq 2\mu e r_{\mathsf{a}} t} P_t(b,c) \\ &\leq t \cdot \exp(-Rt) + \exp(Q(\beta, 2\mu e r_{\mathsf{a}}) \cdot t) \leq (t+1) \cdot \exp(-R't)\end{aligned} \tag{19}$$

for some $R' > 0$, where the last inequality leverages that we know from the proof of Lemma 4 that $Q(\beta, 2\mu e r_{\mathsf{a}})$ is upper-bounded by a negative constant for any $\beta < 2\mu e r_{\mathsf{a}}$. Namely, if $\beta \leq 2\mu r_{\mathsf{a}}$ then we prove in Case 1 that for each β the quantity $Q(\beta, \gamma)$ is maximized by selecting $\gamma = 2\mu r_{\mathsf{a}}$ and this maximum is negative. Likewise, for $2\mu r_{\mathsf{a}} < \beta < 2\mu e r_{\mathsf{a}}$ we show in Case 2 that $Q(\beta, 2\mu e r_{\mathsf{a}}) \leq Q(\beta, \beta) < 0$.

On the other hand, if $b \geq 2\mu e r_{\mathsf{a}} t$ then, considering $b = \beta t$, Lemma 5 directly gives us

$$\sum_{c \geq b} P_t(b, c) \leq \exp(Q(\beta, \beta) \cdot t) \leq \exp(-R'' t) \tag{20}$$

for some $R'' > 0$, as again, we have shown in the proof of Lemma 4 that $Q(\beta, \beta)$ is upper-bounded by a negative constant for any $\beta \geq 2\mu r_{\mathsf{a}}$.

Combining equation (16) with Lemma 3 and equations (19) and (20), we obtain that there exists a constant $R^* > 0$ such that

$$\begin{aligned}
\mathbb{E}[\mathsf{os}(W)^\mu] &\leq L^{\mu-1} \cdot A(\mu) \cdot \sum_{t=0}^{L} \sum_{b=0}^{t} \sum_{c \geq b} P_t(b, c) \\
&\leq L^{\mu-1} \cdot (r_{\mathsf{h}} + r_{\mathsf{h}}^2)^{\mu/2} \cdot \sum_{t=0}^{L} \left(\sum_{b=0}^{2\mu e r_{\mathsf{a}} t} \sum_{c \geq b} P_t(b, c) + \sum_{b = 2\mu e r_{\mathsf{a}} t}^{t} \sum_{c \geq b} P_t(b, c) \right) \\
&\leq L^{\mu-1} \cdot (r_{\mathsf{h}} + r_{\mathsf{h}}^2)^{\mu/2} \cdot \sum_{t=0}^{L} \left(\sum_{b=0}^{2\mu e r_{\mathsf{a}} t} (L+1) \cdot \exp(-R' t) + \sum_{b = 2\mu e r_{\mathsf{a}} t}^{t} \exp(-R'' t) \right) \\
&\leq L^{\mu-1} \cdot (r_{\mathsf{h}} + r_{\mathsf{h}}^2)^{\mu/2} ((L+1)^2 + (L+1)) \sum_{t=0}^{L} \exp(-R^* t) \leq p_1(\lambda)
\end{aligned}$$

for some polynomial $p_1(.)$, as L is clearly polynomial in λ. Returning to (12), we hence obtain

$$\mathbb{E}[\mathcal{G}(\mathbf{B}_{\mathsf{OP}})^\mu] = \mathbb{E}\left[\left(\sum_{i=2L/3}^{L} \mathsf{os}(W_{1:i}) \right)^\mu \right] \leq p(\lambda), \tag{21}$$

again for some polynomial $p(.)$, where the final step follows for example via Minkowski inequality for the norm on the space of random variables defined as $\|Y\|_m := \left(\mathbb{E}\left[|Y|^m\right]\right)^{1/m}$.

5.9 Concluding Security of the Praos Beacon

Invoking Fact 1 with (21) in mind and using $p \triangleq p(\lambda)$, we hence get

$$\Pr[\mathcal{G}(\mathbf{B}_{\mathsf{OP}}) \geq \mathit{\Gamma}] \leq \frac{\mathbb{E}[\mathcal{G}(\mathbf{B}_{\mathsf{OP}})^\mu]}{\mathit{\Gamma}^\mu} \leq \frac{p}{\mathit{\Gamma}^\mu}, \tag{22}$$

where we have the freedom to optimize $\mathit{\Gamma} > 0$.

Recall that we intend to invoke Theorem 1 where the error probability is of the form $k\mathit{\Gamma}\gamma + k\mathit{\Gamma}\epsilon$. Here k is the number of epochs we want to analyze, ϵ is the natural failure rate of $\mathbf{B}_{\mathsf{OP}}$, and $\gamma = p/\mathit{\Gamma}^\mu$ based on (22). We hence optimize $\mathit{\Gamma}$ to minimize this error probability, which occurs when $k\mathit{\Gamma}\gamma = k\mathit{\Gamma}\epsilon$, giving us $\gamma = \epsilon$ and hence $\mathit{\Gamma} = (p/\epsilon)^{1/\mu}$.

Instantiating (22) with these $\mathit{\Gamma}$ and γ results in $\Pr[\mathcal{G}(\mathbf{B}_{\mathsf{OP}}) \geq (p/\epsilon)^{1/\mu}] \leq \epsilon$, and we can use this as the precondition for invoking Theorem 1. We conclude that in the random experiment $X \blacktriangleright_k^{\mathcal{A}} \mathbf{B}_{\mathsf{OP}}$ modeling a k-epoch execution of the protocol $\mathit{\Pi}_{\mathsf{OP}}$, there exists an event E such that

$$\Pr[E] \leq k\mathit{\Gamma}\gamma + k\mathit{\Gamma}\epsilon = 2kp^{1/\mu}\epsilon^{\frac{\mu-1}{\mu}}$$

and moreover, conditioning on $\overline{E}$, the min-entropy of the k-th output X_k of the randomness beacon $\mathbf{B}_{\mathsf{OP}}$ is lower-bounded by

$$H_\infty\left(X_k \mid \overline{E}\right) \geq \lambda - \log \Gamma = \lambda - \frac{1}{\mu} \log\left(\frac{p}{\epsilon}\right) ,$$

even when conditioning on prior beacon outputs. Finally, recall from Proposition 1 that $\epsilon = \epsilon(\mathbf{B}_{\mathsf{OP}}) = \exp(-\Theta(\lambda))$. This allows us to conclude the following theorem. Again, recall Table 1 to interpret the assumption (23) for the case of Cardano.

Theorem 3 (Security of the Praos iterated beacon). *Let $\mathbf{B}_{\mathsf{OP}}$ be the Praos beacon with security parameter λ, denote $\epsilon := \epsilon(\mathbf{B}_{\mathsf{OP}})$, let $\mu \in (1, 2]$ and assume*

$$p_{\mathsf{h}} \geq (11.69\mu - 4.78) \cdot r_{\mathsf{a}} . \tag{23}$$

Consider the random experiment $X_0 \blacktriangleright_k^{\mathcal{A}} \mathbf{B}_{\mathsf{OP}}$ of k-wise sequential composition of $\mathbf{B}_{\mathsf{OP}}$ with any (information-theoretic) adversary $\mathcal{A}$, where X_0 is uniform in $\{0,1\}^\lambda$. Then there exists $p = \mathsf{poly}(\lambda)$ and an event E such that:

1. *The probability that E occurs in the random experiment $X_0 \blacktriangleright_k^{\mathcal{A}} \mathbf{B}_{\mathsf{OP}}$ is at most $2kp^{1/\mu}\epsilon^{\frac{\mu-1}{\mu}}$.*
2. *The output X_k of the beacon's k-th invocation satisfies*

$$H_\infty\left(X_k \mid X_0, \ldots, X_{k-1}, \overline{E}\right) \geq \lambda - \frac{1}{\mu} \log\left(\frac{p}{\epsilon}\right) ,$$

even when conditioning on prior beacon outputs.

Notice that any fixed $\mu \in (1, 2]$ results in a fixed polynomial blow-up of the error ϵ in the error probability appearing in claim 1 of Theorem 3. Therefore, this final error probability is negligible in λ for any fixed $\mu > 1$ and $k = \mathsf{poly}(\lambda)$. With $\mu \to 1$, the condition (23) approaches $p_{\mathsf{h}} \geq 6.91 \cdot r_{\mathsf{a}}$, resulting in $\alpha < 12.4\%$ for the Cardano parametrization (cf. Table 1). Hence, the above analysis in particular implies the following corollary.

Corollary 2. *Assume that $r_{\mathsf{h}} + r_{\mathsf{a}} = 1/20$ as in Cardano, and the adversarial stake share $\alpha < 12.4\%$. In this regime, the Praos family of beacons $(\mathbf{B}_{\mathsf{OP}})_{\lambda \in \mathbb{N}}$ is secure under composition.*

5.10 Grinding under High Observed Chain Density

We now investigate the special case where an additional piece of information is available: the publicly visible chain is *dense* in the sense that it contains at least as many blocks as are *expected to be produced by honest parties*. Namely, the chain created over a sufficiently long interval of length ℓ contains at least $p_{\mathsf{h}}\ell$ blocks. Note that this can happen for several reasons: the honest parties might be lucky enough to win at least the expected number of lotteries (and there is no honest forking), or the adversary contributed to the growth of the public chain, or—and this is the most common situation in practice—the adversary is in fact weaker than the worst-case analysis assumes; for instance, there is no attack currently underway.

The density assumption translates into our analysis as an additional constraint $b \geq p_{\mathsf{h}}t$ and hence $\beta \geq p_{\mathsf{h}}$. Going over the analysis in Sections 5.8–5.9 and in particular recalling the proof of Lemma 4, we observe that it now reduces to only handling Case 3, which leads to a more permissive assumption on the adversarial power (as evidenced by Figure 2). Namely, we obtain the following strengthening of Lemma 4 for the dense case.

Lemma 6. *Let $\mu \in [1, 2]$ and assume that $p_{\mathsf{h}} \geq 2^\mu e r_{\mathsf{a}}$. Then there exists a constant $R > 0$ depending on $p_{\mathsf{h}}, r_{\mathsf{a}}$ such that for any $t \in \mathbb{N}$, $b \in \{\lceil p_{\mathsf{h}}t \rceil, \ldots, t\}$ and $c \geq b$ we have $B_t(b) \cdot C_t(c) \cdot T(b,c)^2 \leq \exp(-Rt)$.*

The rest of the analysis proceeds as in the general case. Formally, the restriction $b \geq p_{\mathsf{h}}t$ defines a new, modified grindable beacon $\mathbf{B}_{\mathsf{OP}}^{\mathrm{d}}$ in which the respective $\mathcal{G}(\mathbf{B}_{\mathsf{OP}}^{\mathrm{d}})$ is determined by the number of option sequences conditioning on $b \geq p_{\mathsf{h}}t$. For this case, we can conclude the following statement limiting grinding against stronger adversaries if a dense chain is observed.

Theorem 4 (Security of dense-chain Praos with iterated beacon). *Consider the modified beacon $\mathbf{B}^d_{\mathsf{OP}}$ obtained from $\mathbf{B}_{\mathsf{OP}}$ by assuming that $b \geq p_{\mathsf{h}}t$. Then the conclusions of Theorem 3 hold for $\mathbf{B}^d_{\mathsf{OP}}$ even when the assumption (23) is replaced by a weaker assumption $p_{\mathsf{h}} \geq 2^{\mu} er_{\mathsf{a}}$.*

Notice that it is sufficient to require the density assumption to hold over intervals of length at least $\log(\lambda)$, as shorter intervals may contribute by at most a polynomial (in λ) number of grinding options.

Chain Density in Cardano. We have collected statistics (as of September 2025, up to epoch 579) about the actual Cardano deployment to empirically support our density assumption. These are summarized in the table below. For each of the three studied time horizons (past 1 month, 1 year, 2 years), we have found the Cardano epoch within that period that contains the least number of blocks in the finalized chain. We recorded the absolute number of blocks, as well as the fraction this number represents out of the 21 600 blocks that are *expected* to appear in a single epoch under full and honest participation.

Timeframe	Min Blocks	Epoch Number	% of 21 600 Blocks
Past 1 month	21 275	575	98.5 %
Past 1 year	20 783	523	96.2 %
Past 2 years	20 404	463	94.5 %

Notice that the empirically observed density of $\geq 94\%$ is sufficient for our argument, as we aim for a security statement against adversaries satisfying $p_{\mathsf{h}} \geq 2er_{\mathsf{a}}$ which translates to an adversarial stake share $\alpha < 15.25\%$ (see Table 1).

Interpreting Theorem 4 for practice. Based on Theorem 4 one can conclude that even adversaries controlling up to about 15% of stake cannot significantly bias the Cardano iterated beacon's output, as long as the produced chain is observed to be dense. Of course, past and present density observations do not guarantee that the density requirement remains satisfied in the (near) future as well. Nonetheless, Theorem 4 can still be operationally leveraged by a Cardano client, as currently observed high chain density puts a bound on *current* grinding, which in turn allows a client to justify—for instance—a more optimistic settlement rule for the upcoming epoch, as blocks are guaranteed to settle faster if the leadership lottery is known to be unaffected by grinding. Other aspects, such as trust in a beacon-based committee selection process exhibit similar dynamics, where an observable evidence of bounded present grinding justifies the expectation of stronger security guarantees in the near future.

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6 Transitioning Praos Analysis to Bounded Message Delays

Our model above adopts a simple, lockstep synchronous networking assumption. However, the Ouroboros Praos protocol has been fully analyzed in a more realistic model reflecting message delays of up to Δ time; in particular, at the adversary's discretion honest messages may be subject to arbitrary delays within time Δ (whereas adversarial messages may of course be delivered to honest parties with no time constraints). Quite precise, concrete security bounds have been obtained for this setting [10], which suggest the possibility of establishing control of grinding with this networking layer. We outline below how the methods above can be adapted to this framework.

Examining the notion of option sequences (Def. 7), we observe that the only role of the honestly produced blocks is to guarantee a length lower bound on the shortest chain that must have been observed by all honest parties by the end of an epoch. In the simple lockstep synchronous setting, chain growth (in the view of any party) grows by at least one for each honest success, and this gives a straightforward lower bound on the length of the longest chain observed by honest participants at the end of an epoch (see (os4) in Def. 7).

In the setting with Δ-delay, the rate of chain growth is known [9,10] to correspond to the effective "honest depth" $h_\Delta(x)$ of the characteristic string x. This quantity is recursively defined so that $h_\Delta(\epsilon) = 1$ (where ϵ is the empty characteristic string of length 0), $h_\Delta(xa) = h_\Delta(x)$ if x is a characteristic string and a is a (characteristic string) symbol with no honest successes, and $h_\Delta(xa) = h_\Delta(\mathrm{trim}(x)) + 1$ if a is a symbol with an honest success—here $\mathrm{trim}(x)$ denotes the string obtained by removing the last Δ symbols of x. It is an easy exercise to determine the expected value of $h_\Delta(x)$ when x is drawn from the distribution $\mathcal{L}_n(r_\mathsf{h}, r_\mathsf{a})$: If

$p_{\mathsf{h}} = \Pr[h_i > 0]$ (where $w_i = (h_i, a_i)$), then the expected value is $n/\alpha + O(1)$, where $\alpha = \Delta + (1 - p_{\mathsf{h}})/p_{\mathsf{h}}$ [9, Lemma 15].

The major challenge is that one then needs very precise tail bounds for the quantity $h_\Delta(w_1, \ldots, w_n)$ around this mean. Generic approaches (e.g., the McDiarmid inequality used in [9] for this purpose), will not suffice because the constant appearing in the exponential decay terms (analogous to (17)) is critical. To handle this, consider a long characteristic string $w_1, w_2, \ldots$ and consider the dual random variable T_d equal to the smallest t for which $h_\Delta(w_1 \ldots, w_t) = d$; this random variable has the pleasant decomposition $T_d = T_{d-1} + \Delta + G$, where G is a geometric random variable with parameter p. This is explained by the fact that a height increase is followed by a Δ period in which no height increase can occur after which the next honest success induces a height increase. Thus $T_d = d\Delta + (G_1 + \ldots + G_d)$ and we can rely on high-precision tail bounds for sums of geometric random variables to establish the necessary bounds [11]. Observe that, as it must, this will determine a threshold that depends on Δ as a parameter. Even without determining the constants here, we remark that this demonstrates that even in this model, *some* constant fraction of adversarial stake is necessary to launch a successful attack on the Cardano blockchain, even with unlimited resources allocated to grinding.